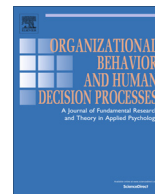




Contents lists available at ScienceDirect

Organizational Behavior and Human Decision Processes

journal homepage: www.elsevier.com/locate/obhdp

The utility of a multifoci approach to the study of organizational justice: A meta-analytic investigation into the consideration of normative rules, moral accountability, bandwidth-fidelity, and social exchange

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ARTICLE INFO

Article history:

Received 28 June 2012

Accepted 23 October 2013

Accepted by Maureen Ambrose, Scott Reynolds and Marshall Schminke

Keywords:

Ethics

Justice

Fairness

Social exchange

Multifoci justice

Target similarity

Behavioral ethics

Organizational justice

Meta-analysis

Accountability

ABSTRACT

Multifoci justice pulls from research on social exchange theory to argue that despite the proliferation of rule sets in the literature (often referred to as the “types” of justice), individuals seek to hold some party accountable for the violation/upholding of such rules, and it is these parties (e.g., supervisors, the organization as a whole) that are most likely to be the recipients of attitudes and behaviors (i.e., target similarity effects). To explore these issues, we meta-analytically ($k = 647$, $N = 235,682$) compared the predictive validities of source- vs. type-based justice perceptions and found that (a) multifoci justice perceptions more strongly predicted outcomes directed at matched sources than did type-based justice perceptions, (b) multifoci justice perceptions more strongly predicted target similar than dissimilar outcomes, and (c) the relationships between multifoci justice perceptions and target similar outcomes were mediated by source-specific social exchange.

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“One’s sense of fairness is grounded in basic ethical assumptions of normative treatment. The sense of injustice, therefore, often involves holding someone accountable for a deliberate transgression of acceptable conduct”

- Folger, Cropanzano, & Goldman, 2005, p. 216

“Justice types, although distinct, may not in the final analysis be very different in either their dynamics or their consequences.”

- Lind, 2001, p. 225

Introduction

The fields of organizational justice and behavioral ethics are inextricably linked via their joint focus on ethics-related judgment

and decision making processes. Whereas behavioral ethics research often focuses on individuals’ conformity to rules and norms of morality and ethicality (e.g., Treviño, Weaver, & Reynolds, 2006), or the extent to which moral judgment enters into decision processes (e.g., Loviscky, Treviño, & Jacobs, 2007; Thoma & Rest, 1999), organizational justice research largely focuses on how individuals judge the actions of others and how these judgments drive subsequent attitudes and behaviors, also focusing on individuals’ use of normative rules for what is considered ethical or just (Folger, 2001; Singer, 2000). A fundamental phenomenon shared by both of these fields is moral accountability (Ciulla, 2004; Folger & Cropanzano, 2001). That is, regardless of whether the focus is on actors or observers, or on victims or perpetrators, the judgments studied in both of these domains require actions for which there is a responsible party (Petrick & Quinn, 2001; Smiley, 2009).

In this paper, we explore two distinct but overlapping aspects of justice perception formation: normative rules and moral or ethical³

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³ Consistent with the behavioral ethics literature (Treviño et al., 2006), we will use the terms morality and ethics interchangeably in this paper.

accountability. The former operationalizes justice perceptions as individuals' assessment of the extent to which normative rules (i.e., distributive, procedural, and interactional justice rules) have been upheld or violated. The latter focuses on how specific parties (e.g., supervisors, coworkers, the organization as a whole) are seen as upholding or violating these normative rules, as well as the relational processes that follow such judgments. Although these differences may seem subtle at first glance, and although the perspectives often overlap, in this paper we seek to investigate just how important precision with regard to these two characteristics is for understanding how individuals react to workplace (in)justice.

In the following sections, we review the theoretical foundations of normative justice rules and accountability processes. Then, based on research on bandwidth-fidelity (i.e., the alignment of the specificity of predictor and criterion variables), moral accountability, and social exchange, we offer three hypotheses suggesting that approaches that incorporate *both* notions of normative rule sets and accountability processes will have the most utility for predicting variance in attitudinal and behavioral outcomes. To test these hypotheses, we coded the scales used to measure justice variables across studies spanning nearly 30 years (over 640 studies involving over 235,000 participants) according to these two characteristics. This allowed us to use meta-analysis combined with regression and path analysis to consider the relevance of justice rules and accountable parties in explaining variance in outcomes targeting such parties, as well as the mediating role of social exchange in such processes.

Justice “types”: the application of normative rules

Distributive, procedural, and interactional justice (DJ, PJ, and IJ, respectively) are often written about and described as “types” of justice perceptions (Cohen-Charash & Spector, 2001; Colquitt & Shaw, 2005). That is, the constructs are assumed to represent employees' perceptions of whether the outcomes they receive, the procedures used to determine outcomes, and both the information conveyed about procedures and the interpersonal treatment bestowed as information is shared, are all *fair*. In this sense, the perceptual targets of justice judgments are thought to be outcomes, procedures, and informational/interpersonal encounters, and as a result, for decades, justice has been treated as a multidimensional construct, with these justice types modeled as predictors of workplace outcomes (for reviews, see Cohen-Charash & Spector, 2001; Colquitt, Conlon, Wesson, Porter, & Ng, 2001).

Despite being commonly treated this way, outcomes, procedures, and interactions were not originally theorized as perceptual targets, but rather as labels for sets of rules or criteria used by perceivers to *arrive at* justice perceptions. Thus, DJ, PJ, and IJ, were not originally proposed to be fairness perceptions on their own. Rather, they have been theorized to be one piece of the perception formation process whereby a focal party is deemed to be fair or unfair (Lind & Tyler, 1988). Further, this tripartite model was not the product of a unitary theory focused on mapping the perceptual domain of justice. Rather, these three constructs (or four, if interactional justice is further subdivided into informational and interpersonal justice) represent three research traditions developed over a long period of time, which have spanned multiple theoretic disciplines (see Byrne & Cropanzano, 2001).

The lion's share of early justice research focused on the social psychology of *distributive justice*, and largely dealt with the rules applied by employees to determine the fairness of outcomes such as pay. This included rules of equity (Adams, 1965), equality, and need (Deutsch, 1975; Leventhal, 1976). The concept of *procedural justice* was introduced by Thibaut and Walker (1975), who proposed process control, and Leventhal (1980), who proposed consistency,

bias-suppression, accuracy, correctability, representativeness, and ethicality as the normative criteria for procedural justice. This work took a social psychological approach, but was often couched in the legal literature and focused on legal phenomena (Thibaut & Walker, 1978; Thibaut, Walker, La Tour, & Houlden, 1974). The intention here was to show that procedural justice can mitigate the backlash caused by delivering unfavorable outcomes. Much later, organizational scholars proposed that additional rules seemed to govern the acceptability of how procedures were implemented in organizations, and added politeness, dignity, respect, and the thoroughness of explanations (delivered by the party accountable for carrying out the procedure) as criteria for *interactional justice* (Bies & Moag, 1986; Greenberg, 1993).

Thus, from multiple disciplines and across four decades emerged three sets of criteria used *as part of* the justice perception formation process. Each rule set was proposed in a sequential fashion, and was argued to moderate the effects of the dominant construct of the time. DJ effects were shown to be mitigated by PJ, PJ effects were shown to be mitigated by IJ, and so forth (see Brockner, 2010 for a review of the vast literature on DJ $\times$ PJ $\times$ IJ interactions; and Cropanzano & Ambrose, 2001 for discussions of the interchangeability of justice types; see also Van den Bos, Lind, Vermunt, & Wilke, 1997). What we feel is important to note is that somewhere along the line, researchers took this largely contextual research (i.e., much of this research treated justice perceptions not as perceptions in the attitudinal sense, but as self-reports of whether management complied with normative standards surrounding outcome favorability, organizational policies, and interpersonal treatment) to represent the classes of judgments made by employees. This led to the development of well-validated measures of these three (four) facets, and meta-analyses which many saw as confirming this perceptual structure (Colquitt, 2001; Colquitt et al., 2001). Yet Colquitt and colleagues, who are by far the most cited supporters of this structure, have made clear in their writing that the type-based justice measures are nothing more than employees' reports on whether or not the normative justice rules posed in past theories have been violated (with each item largely representing one of the rules or criteria).

Ambrose and Arnaud (2005) also pointed out that these rules are technically antecedents to the justice types, and thus not the elements of justice perceptions *per se*. Thus procedures, outcomes, and interpersonal treatment may not constitute perceptual targets, but rather theory-based sets of rules that employees can use to evaluate their working conditions, which are likely used in the process of holding a party accountable for their actions (see also Lind, 2001). Whereas these rules have formed the zeitgeist of justice scholarship, and whereas research has long argued that individuals are particularly sensitive to violations of normative rules surrounding the treatment of others (Mikula, Petri, & Tanzer, 1990; Treviño & Bies, 1997), the specification of differential parties, differentially held accountable for different types of justice violations, has received considerably less attention in comparison. We find this surprising given the universal agreement regarding the importance of accountability in how individuals come to perceive and react to injustice (Cropanzano, Chrobot-Mason, Rupp, & Prehar, 2004; Folger & Cropanzano, 2001).

Holding parties morally accountable for normative rule violations

The concept of moral accountability has been a prevailing theme throughout the history of justice scholarship, and remains a salient feature of behavioral ethics research (Weiner, 1995). For example, ethics research has uncovered the importance of blame attributions for predicting responses to unfair behavior (e.g., revenge, reconciliation, forgiveness; Aquino, Tripp, & Bies, 2001; Bradfield & Aquino, 1999). Much of this research pulls from

earlier work in the areas of conflict management and dispute resolution, which explores the process whereby grievances are transformed into disputes. According to this research, disputes occur when an injurious offense has occurred, and the harmed party holds another party responsible for the injury (i.e., a blame attribution; Felstiner, Abel, & Sarat, 1980–1981; Ferguson & Rule, 1983).

Fairness theory. Such a process is similar to that proposed by fairness theory (Folger & Cropanzano, 2001; Folger et al., 2005), in which justice perceptions are thought to be the product of an accountability process. According to this theory, at the point at which harm is inflicted on another, victims (or observers) determine (a) if the harm was the outcome of another's (avoidable) actions, and (b) if the accountable party's actions violated normative standards of moral behavior. Only if these conditions are met is the victim or observer expected to perceive an injustice as having occurred, and only in meeting these conditions are attitudinal and behavioral responses expected that particularly target the source of the injustice. In fact, and consistent with deontological ethics (Broad, 1930; Kant, 1785/2002), Greenbaum, Folger, and Ford (2011) have argued that it is really only such targeted reactions to perceived violations of moral contracts that can be truly classified as a justice response. This leads to the argument that perceptions of normative justice rule violations (i.e., general justice “type” perceptions), without consideration of an accountable party to which outcomes are directed, while distinguishable and predictive of general attitudes and behaviors, may not be isolating justice reactions completely, and in the way implied by the literature.

In addition, as Folger and Cropanzano (2001) note, there is a high likelihood that our common sets of normative justice rules, when not specifically tied to an accountable party, may not only be attributionally ambiguous (making the assignment of blame, and consequently, the formation of true justice perceptions difficult), but also that the different types of justice may vary in their level of attribution ambiguity. This perspective also argues that individuals are capable of holding different parties accountable for moral violations—including both individual transgressors as well as organizations as juristic persons (Coleman, 1982, 1993)—and that just as differential treatment could come from different parties, individuals' responses to the behaviors of these different parties would be expected to be accordingly differentiated.

Interestingly, although moral accountability and attributions of blame are common topics within the behavioral ethics literature; the justice literature, despite deep theoretical roots to notions of accountability, has been slower in explicitly specifying how unique sources of justice (i.e., distinct transgressors) can differentially impact the relationships, attitudes, and behaviors of those impacted by their actions.

Multifoci justice. An exception to this is the multifoci approach (Cropanzano, Byrne, Bobocel, & Rupp, 2001). The multifoci perspective argues that since employees work within complex social networks, there exist myriad parties that employees evaluate and react to through the lens of justice. Multiple sources or *foci* of justice have been studied within this area, including supervisors (Masterson, Lewis, Goldman, & Taylor, 2000), the organization as a whole (Cobb, Vest, & Hills, 1997; Colquitt & Shaw, 2005; Rupp & Cropanzano, 2002), co-workers (Donovan, Drasgow, & Munson, 1998), and customers (Skarlicki, Van Jaarsveld, & Walker, 2008; Wang, Liao, Zhan, & Shi, 2011). The multifoci perspective posits that asking employees about the extent to which normative standards regarding the distribution of outcomes, the

enactment of procedures, or the delivery of interpersonal treatment are upheld, without reference to the particular party held responsible for the fairness of such matters, is missing an important piece of the phenomenon. This perspective takes umbrage to excluding a “face” from justice – a construct that is theoretically rooted in the notion of accountability, and therefore requiring of a responsible party (Cropanzano et al., 2004; Folger & Cropanzano, 2001).

Target similarity: aligning sources of justice and targets of outcomes theoretically (via accountability and social exchange) and methodologically (via bandwidth-fidelity)

Both the ethics literature on blame attributions and revenge (Aquino et al., 2001; Bradfield & Aquino, 1999), as well as the research on fairness theory (Folger & Cropanzano, 2001) speak very specifically about individuals' targeted reactions toward transgressors deemed accountable for unethical or unfair acts. The multifoci perspective has this same theme (Horvath & Andrews, 2007; Karriker & Williams, 2009). That is, justice perceptions made about a particular party (i.e., perceptions that an accountable party has violated normative justice rules) are expected to be most proximately related to attitudinal and behavioral reactions directed specifically at *that party*. Indeed, the alignment of justice source with response target has been termed “target similarity” by Lavelle, Rupp, and Brockner (2007; see also Lavelle, Rupp, Manegold, & Thornton, in press). Thus, throughout the paper, we use the term *multifoci justice* to refer to justice perceptions made about particular parties, and *target similarity* to refer to the effect of multifoci justice perceptions on outcomes directed at the same source.

On one hand, aligning the source of justice perceptions with the target of attitudinal and behavioral reactions and expecting better prediction of outcomes might be seen more as a methodological artifact than a theory-based phenomenon, in that decades of research shows that better prediction of outcomes can be expected when the level of specificity of the predictor matches that of the criterion. For example, the theory of reasoned action (Ajzen & Fishbein, 1977; Fishbein & Ajzen, 1975) was a response to a long history of research showing weak associations between attitudes and behaviors. More precision was found in the predictive validity of attitudes not only when behavioral intentions were modeled as more proximal attitudinal outcomes, but also when the target of the attitude matched the target of the behavioral intention. That is, rather than predicting a global behavior (e.g., health) with a specific attitude (attitude toward smoking), one is better off predicting behaviors that are at a similar level of specificity to the attitude in question (e.g., smoking behavior). Similar arguments emerge from the concept of bandwidth-fidelity (Cronbach, 1970; Cronbach & Gleser, 1965). This tradition has shown undisputedly that prediction is optimized when predictor and criterion specificity are matched. Such issues have been deeply discussed with regard to personality traits (vs. facets) predicting job performance (vs. specific performance dimensions; Hogan & Roberts, 1996; Ones & Viswesvaran, 1996), as well as with regard to whether measures of counterproductive work behavior should be general or specific (Bowling & Gruys, 2010). Even within the justice domain, specificity matching has been advocated for (Colquitt & Shaw, 2005). In fact, Ambrose and Schminke's (2009) proposal of the overall justice construct was predicated on the fact that the outcome variables of interest to justice researchers were so general (e.g., citizenship, deviance, performance), that a more general justice construct would likely be a more proximal predictor compared to more

specific justice facets (e.g., DJ, PJ, IJ; which they confirmed empirically). Certainly, the “matching” advocated for by the target similarity perspective conveys a similar tone, by arguing that in studying justice phenomena, the sources of justice and targets of outcomes should be aligned.⁴

On the other hand, whereas that might certainly explain some of the incremental validity to be theoretically gained by adding precision to one's model with regard to justice source and outcome target, the target similarity perspective argues that it is more than just this. That is, multifoci research is grounded in social exchange theory (Blau, 1964; Emerson, 1962, 1972; Homans, 1961), which articulates a psychological process by which norms of reciprocity generate relational capital between employees and those parties with whom they interact. Considerable research has shown evidence for such reciprocity-based (and hence target-specific) response patterns. For example, research has shown that perceived organizational support engenders organization-directed commitment, whereas leader-member exchange engenders supervisor-directed citizenship behavior (Settoon, Bennett, & Liden, 1996). A key aspect of social exchange theory (as applied to justice theory) is that *fairness drives social exchange*. Hence, when a specific party treats employees fairly, social exchange is engendered, which generates positive reactions directed back at the exchange partner (Cropanzano & Mitchell, 2005; Lavelle et al., 2007). In sum, the multifoci perspective integrates tenants of behavioral ethics and fairness theory pertaining to holding parties accountable for the upholding/violations of normative rules of fairness, and adds to this a mediating role of source-specific social exchange in influencing reactions to source-specific justice perceptions. At the same time, this research is consistent with what social psychology and the research methods literatures have argued about aligning variables in terms of their scope and specificity.

Support for the multifoci model. At its start, multifoci research made assumptions about who specifically employees hold accountable for the different types of justice. That is, early research proposed that supervisors were the implied focal party responsible for interactional justice (IJ), whereas organizations were the implied focal party responsible for procedural justice (PJ). It was expected that interactional justice would engender social exchange relationships with supervisors (e.g., leader-member exchange) and consequently impact outcomes directed toward supervisors (e.g., supervisor-directed citizenship behaviors), whereas procedural justice would engender social exchange relationships with the organization (e.g., perceived organizational support) and consequently impact outcomes directed toward the organization (e.g., organization-directed citizenship behaviors; Moorman, Blakely, & Niehoff, 1998; Roch & Shanock, 2006). Some research has supported these predictions (see Cropanzano, Prehar, & Chen, 2002; Fassina, Jones, & Uggerslev, 2008; Jones, 2009; Malatesta & Byrne, 1997; Masterson et al., 2000; Stinglhamber, De Cremer, & Mercken, 2006), although as we will point out below, this assumption has not been completely borne out in the extant literature.

Subsequent research has argued against the notion that employees will *automatically* hold a particular party accountable for a particular type of justice. Instead, in the modern workplace, supervisors, organizations, coworkers, etc., might all possess the capability to deliver outcomes, processes, and interpersonal treatment and as such, a wider array of justice perceptions should be

expected. Some discussion of the source of justice dates back to the early nineties, shortly after the construct of interactional justice was introduced in the literature. During this time, researchers were debating whether or not interactional justice was simply a social aspect of procedural justice. Tyler and Bies (1990) system-agent model argued that agents (e.g., supervisors) have as much of an impact on perceptions of procedural justice as the system (e.g., the organization). Cobb et al. (1997) found that employees receiving performance appraisals were able to make distinctions between the fairness of the organization's formal policies and the fairness of policies set by supervisors. Byrne (1999) was the first to find support for a “fully-crossed” model, consisting of supervisory PJ, supervisory IJ, organizational PJ, and organizational IJ. Rupp and Cropanzano (2002) replicated this model, and showed that (consistent with a target similarity perspective) social exchange relationships with each party mediated the effects of justice perceptions made about a party on outcomes directed at that party (see also Frazier, Johnson, Gavin, Gooty, & Snow, 2010). Olkkonen and Lipponen (2006) found similar effects, and extended the model to also include organizational DJ. Further replication and expansion was provided by Karriker and Williams (2009), who also considered multifoci DJ.

Extending these ideas further, Lavelle et al. (2009) showed across two studies that one's workgroup can be an important source of justice. They showed that organizationally-focused procedural justice predicted organization-directed citizenship, mediated by organizational commitment (note that commitment has been argued to be a proxy for social exchange, as have variables such as trust, support, and identification; see Cropanzano & Byrne, 2000; Lavelle et al., 2007). They further showed workgroup-focused procedural justice predicted individual-directed citizenship, mediated by workgroup commitment. Other research has focused on customers as another source of justice, with perceptions of customer fairness impacting customer-directed attitudes and behaviors (Rupp, McCance, Spencer, & Sonntag, 2008; Rupp & Spencer, 2006; Skarlicki et al., 2008; Spencer & Rupp, 2009).

Hypotheses

Although this collection of empirical research shows promise for the multifoci perspective, the model and its value are yet to become widely recognized within the broader literature, and justice researchers still largely rely on the traditional type-based factor structure of justice to conceptualize and operationalize justice perceptions without explicit, systematic, and consistent consideration of source/accountability in employees' formation of justice perceptions. Indeed, in analyzing the 360+ studies that have come out since the multifoci model emerged (which we trace to Malatesta & Byrne, 1997), we have determined that less than 4% have been explicitly multifoci in their perspective. Further, studies that have incorporated measures of DJ, PJ, and/or IJ have made no reference to source, or made reference to multiple sources in the same measure, nearly 35% of the time. In summary, if there is inferential power to be gained by taking a multifoci approach, by virtue of both modeling the full theoretical domain of justice perceptions (namely, accountable parties held responsible for upholding/violating justice rules), as well via improvements in matching variable specificity and modeling social exchange processes, this has yet to be widely accepted by the research community. Consequently, in this study, we sought to explore the extent to which incremental validity is to be gained by connecting normative justice rules with particular parties held accountable for upholding/violating them, and the extent to which social exchange plays a key role in explaining multifoci effects.

First, we predict that source-based justice perceptions will offer greater utility in predicting source-directed outcomes as compared

⁴ Note however, that this perspective differs from how this argument has played out in the personality and performance literatures, in that target similarity arguments do not always pertain to general vs. specific constructs (e.g., misalignment could also result from mismatched constructs with the same level of specificity, such as coworker-focused justice perceptions and supervisor-directed outcomes). Rather it surrounds the expectation that responses to injustice will be most strongly directed at those held accountable for unjust acts.

to type-based justice perceptions (which contain reference to normative rules but are imprecise with regard to the accountable party):

Hypothesis 1. Multifoci justice perceptions more strongly predict outcomes directed at matched sources than do type-based justice perceptions (i.e., distributive, procedural, and interactional justice perceptions).

Second, we predict that matched (or target similar) multifoci effects will be larger as compared to mismatched, or target dissimilar effects. It is important to note the relative nature of this hypothesis. Whereas we expect target similar effects to be larger than target dissimilar effects, this does not suggest that we expect an absence of target dissimilar effects. Lavelle et al. (2007) speak quite explicitly about the possibility of cross-foci effects between sources. This is especially the case for supervisory-focused justice perceptions predicting outcomes targeted at other sources. Such an expectation is based on research pointing to the poignant role of the supervisor in shaping the attitudes and behaviors of employees, such that reactions to supervisory treatment can often “spread” to other targets, and be more generally dispersed due to its salient impact on the intrapsychic processes of employees (e.g., Masterson et al., 2000; Olkkonen & Lipponen, 2006; Rupp & Cropanzano, 2002; Stinglhamber & De Cremer, 2008).

What they specify, however, is that target dissimilar effects will be smaller in magnitude (on average) as compared to target similar effects. Given that employees are likely to experience (un)fairness from multiple parties simultaneously at work (supervisors, organization, co-workers, customers), we expect that the target similarity effects for multiple justice sources will hold simultaneously in the presence of potential cross-foci effects (which are expected to be smaller in magnitude).

Hypothesis 2. Multifoci justice perceptions more strongly predict outcomes directed at matched (vs. mismatched or target dissimilar) sources.

Finally, whereas we acknowledge that issues of bandwidth and specificity play a role in the incremental predictive validity to be gained by modeling multifoci justice constructs as predictors of matched, target-similar outcomes, we also hypothesize social exchange as a unique mediating mechanism undergirding source-based justice phenomena. We seek to provide evidence that this phenomenon is more than a methodological artifact, by showing that multifoci direct effects are attenuated when source-specific social exchange is added to our theoretical model.

Hypothesis 3. The relationship between multifoci justice perceptions and matched source-directed outcomes will be mediated by social exchange with the respective source. Such mediation effects will exist simultaneously for multiple sources, and will exist above and beyond possible cross-foci effects.

Conflicting results. By carrying out this study, we also sought clarity with regard to some conflicting results that have emerged very recently in the literature. That is, despite the growing number of studies that have taken a multifoci approach, which have presented different forms of evidence substantiating the target similarity perspective, a meta-analysis was recently published that drew different conclusions than what this extant research has been suggesting. Specifically, Colquitt et al. (2013) carried out a meta-analytic review of the justice literature with the goal of presenting an update to the Colquitt et al. (2001) meta-analysis. Given that a number of trends had developed in the literature since the earlier meta-analysis was published, these authors also sought to explore the potency of a number of theoretical lenses including affect, social exchange, and the relative potency of events vs. global entities. After comparing

the average bivariate correlations between organizationally- and supervisory-focused distributive, procedural, and interactional justice perceptions and target-specific outcomes, based on the overlapping confidence intervals across the majority of these relationships, the authors concluded that matching the source of justice with the target of an outcome does not lead to improved prediction over the traditional, unmatched DJ, PJ, IJ framework.

In contrast to the analytical approach taken by these authors (examining only bivariate correlations), the majority of multifoci justice research to date has (a) examined at least two sources of justice simultaneously; (b) taken a multiple regression or structural equations modeling approach in order to model the combined and incremental effects of multiple sources of justice; and (c) hypothesized and/or modeled cross-foci effects (which are expected to exist but be of a relatively smaller magnitude compared to target similar effects). Therefore, to adequately test the relative strength of the target similarity model meta-analytically, a more sophisticated analytical approach is needed to parse out the incremental effects of source and type (like that taken by Colquitt et al. (2001) in exploring the system-agent model). Thus, a second goal of our research was to further explore the discrepancy between the results reported in the extant multifoci justice research and the conclusions drawn from Colquitt et al. (2013) by applying hierarchical multiple regression and path analysis within a meta-analytic framework.

Summary. In conclusion, we seek to examine whether prediction of source-directed outcomes is optimized by mapping the entire perceptual process outlined in the literature (which involves holding particular parties accountable for the upholding/violation of normative justice rules). We also explore social exchange as a psychological mechanism driving these effects that exerts influence above and beyond the methodological precision gained by modeling target similar predictors and outcomes. We see this approach as bringing theory, measurement, and the full process of justice perception formation into closer alignment.

Method

We tested our hypotheses by looking back at the empirical organizational justice literature, assembling and coding every scale used to measure every variable in the studies identified. Doing so allowed us to determine the relative robustness of type-based justice effects (which are mixed in terms of source) vs. source-based justice effects (which contain types as well as one specific source) on source-directed outcomes. It also allowed us to test the target similarity model meta-analytically, considering the effect sizes of justice-outcome relationships that were matched vs. mismatched in terms of source. Finally, with these data, we were able to test the mediating role of social exchange in explaining multifoci effects meta-analytically.

Our coding revealed that the majority of justice research carried out across the decades has referenced three types of justice: distributive, procedural, and interactional; two sources: supervisors and the organization as a whole; five social exchange constructs: support, trust, commitment, identification, and exchange; and four major outcome variables: job performance, satisfaction, citizenship (including civic virtue, loyalty, and sportsmanship), and counterproductive behaviors (including sabotage, retaliation, theft, aggression, and deviance). Table 1 lists and defines these variables, which were used to test our hypotheses.

Identification and selection of studies

In order to identify prior empirical studies of organizational justice, we conducted a literature search in the PsycINFO, ABI/Inform

(ProQuest), PsycArticles (Ebscohost), JSTOR, Academic Search Complete, Business Source Complete, and ProQuest Dissertations and Theses databases, from the 1800s through 2012 using the search terms: *(in)justice*, *(un)fairness*, *distributive justice*, *procedural justice*, *interactional justice*, *interpersonal justice*, *informational justice*, *organizational justice*, and *mistreatment*. This search produced 1154 possible articles and 332 possible dissertations. This included studies subsumed within both Colquitt et al.'s (2001) and Cohen-Charash and Spector's (2001) meta-analyses, papers that were in these authors' date ranges but not included, as well as papers published following these authors' date ranges (through the year 2012).

Next, we read through the identified articles in order to determine if they would be suitable for our meta-analysis based on multiple selection criteria. First, the article needed to be empirical in nature. Second, the article needed to assess organizational justice (i.e., not more general constructs, such as social justice) and the study needed to be conducted in an organizational (or simulated organizational) setting. Third, we chose to focus on studies that included either a correlation matrix, or provided additional information for calculating at least one zero-order correlation between a justice and an outcome variable. Fourth, due to our focus on social exchange mediating the relationship between perceived justice and targeted outcomes, we only focused on studies examining first-party justice (i.e., employees' personally experienced or perceived justice as opposed to their reports on how fairly others are treated). As a result, studies examining the relationships between third-party justice perceptions and third-party observers' reactions were excluded from our data analyses. Fifth, we only included correlations in our database pertinent to individual-level justice effects. Thus, group- or organizational-level correlations (e.g., justice climate effects) were excluded. Based on these selection criteria, our final meta-analysis database included data from 530 papers (484 empirical articles and 46 dissertations). From these 530 papers, there were 647 distinct studies (due to some papers including multiple studies and/or multiple samples), which in total, contained over 12,236 correlation coefficients among justice and outcomes. The total meta-analytic sample size was 235,682 participants. The list of studies included in the meta-analyses is available at <http://doiop.com/refe2013>.

Data coding

Coding of justice measures

For every paper, we first determined what measures were used for each study. If the full measures were reported in the paper, those items were coded directly. If the full measures were not included, but the study authors stated that no modifications were made to the original scale, we obtained the measure from the paper where the original measure was first introduced, and coded these items. If the full set of items were not included in the original paper either, we contacted the (original) study authors to request the full measure. If the measures were not included, but any aspect of the theory, methods, or results suggested that any of the measures were modified for the purpose of the study, those measures were obtained from the study authors directly. We designed coding sheets with which to capture relevant information pertinent to each study, such as zero-order correlations between variables of interest, measures used, reliabilities of the scales used to assess the variables, sample size, and other information. For each justice measure used in each study, we recorded the justice type and source based on the information gleaned from the paper (and any further investigation that was required).

Type referred to whether outcomes, procedures, or interactions were referenced in the items. Source referred to the party that is referenced in the items as the "deliverer" of fair or unfair treatment

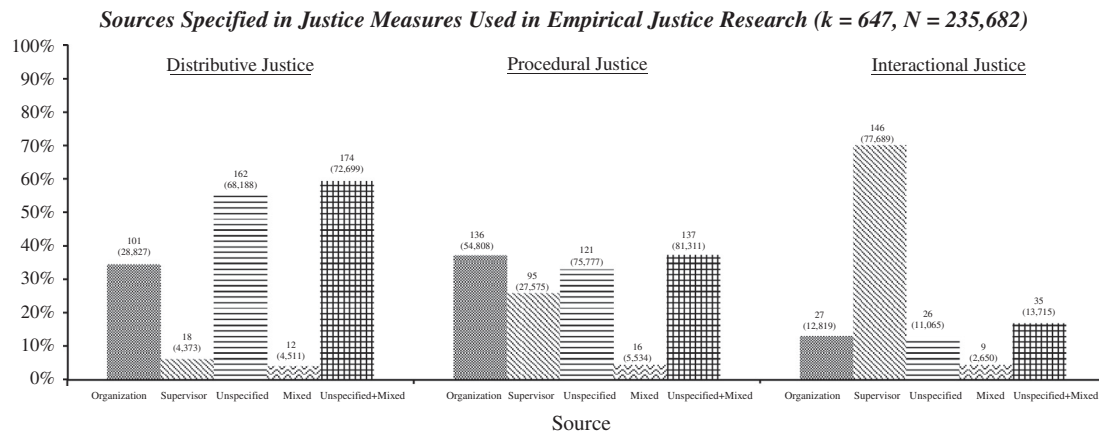
(e.g., organization, supervisor/manager). Whereas type was almost always explicitly labeled, many of the papers were ambiguous in both the hypotheses offered and measure description regarding who was being referenced by the justice items. However, source could often be determined by investigating both the items themselves, as well as authors' theoretical arguments. For example, items used in Karriker and Williams' (2009) study of justice and organizational citizenship behavior all took the form: "My outcomes that are controlled by my supervisor are justified, given my performance." A scale comprising items such as these was coded type = distributive justice, source = supervisor. If the items contained either no reference to a particular source, or referenced multiple sources across the items (e.g., injustice from both the supervisor and the organization), we coded the justice source as "unspecified" or "mixed," respectively. Studies falling within these categories were not included in the analyses of source-based justice perceptions, but were included in the analyses of type-based justice perceptions (if type was specified). As a result of our coding strategy and following the multifoci justice approach, supervisory-focused justice perceptions included distributive, procedural, and interactional fairness perceptions made about one's supervisor, whereas organizationally-focused justice perceptions included the distributive, procedural, and interactional fairness perceptions made about one's organization. Distributive justice perceptions included distributive fairness perceptions made about supervisors, the organization, and from mixed and unspecified sources. The same was true for both procedural and interactional justice (i.e., both, respectively, included perceptions made about supervisors, the organization, and from mixed and unspecified sources).

Fig. 1 describes what the coding process revealed about the presence of type and source referents in the actual measures used in the extant organizational justice literature. As we reviewed above, research on the system-agent model argued for the organization as the source of procedural justice, and the supervisor as the source of interactional justice. Research from this tradition (McFarlin & Sweeney, 1992; Moorman, 1991), in addition to early multifoci research (Malatesta & Byrne, 1997; Masterson et al., 2000) assumed that research participants would assume these referents, whereas later multifoci research advocated for being more explicit with regard to referent (e.g., Rupp & Cropanzano, 2002). Further, Colquitt's measure, which was introduced in 2001, specifically allows for foci of interest to be included in the instructions for completing each subscale. Fig. 1 seems to bear out this evolution. For distributive justice, whereas the organization is referenced more than the supervisor, the majority of studies have used distributive justice measures that did not explicitly specify a source. For procedural justice, the use of an organizational referent, a supervisor referent, or an unspecified or mixed referent has been more evenly distributed across studies. For interactional justice, past research seems to have more consistently referenced the supervisor, with the organization and mixed/unspecified sources being referenced less often. What is consistent with our more anecdotal observations at the start of this paper, is that a significant proportion of the population of measures, while specific in terms of types, were either not specific or inconsistent with regard to the specification of source (40% of all measures). What is important to note for interpreting the tests of our hypotheses, is that for the meta-analytic estimates computed, whereas the multifoci justice estimates contain all three sets of normative justice rules (i.e., distributive, procedural, and interactional—but consistent in their source, i.e., supervisor or organization), the justice types contain conflicting mixes of sources, including justice measured specifying no source at all. This issue gets to the heart of our hypotheses, as to why multifoci approaches might offer more precision with regard to predicting source-directed outcomes, via source-specific social exchange. We return to this issue below.

Table 1

Definitions of variables involved in organizational justice research.

Construct	Definition
<i>Justice</i>	
Supervisory-focused Justice (SJ)	Perceptions of the (un)fairness received from one's supervisor (Lavelle et al., 2007; Rupp, 2011)
Organizationally-focused Justice (OJ)	Perceptions of the (un)fairness received from one's employing organization (Lavelle et al., 2007; Rupp, 2011)
Distributive Justice (DJ)	Perceptions regarding the following of allocation rules (Adams, 1965; Homans, 1961)
Procedural Justice (PJ)	Perceptions regarding the following of the rules of procedural justice (Thibaut & Walker, 1975)
Interactional Justice (IJ)	Perceptions regarding the following of normative rules of interpersonal treatment (Bies & Moag, 1986), which consists of two aspects: (a) <i>interpersonal justice rules</i> —treating one with dignity and respect (Bies & Moag, 1986); and (b) <i>informational justice rules</i> —providing adequate explanation for decisions (Greenberg, 1993)
<i>Organizational Citizenship Behavior (OCB)</i>	
	"Individual behavior that is discretionary, not directly or explicitly recognized by the formal reward system, and that in the aggregate promotes the effective functioning of the organization" (Organ, 1988, p. 4)
Citizenship Behavior targeting the Organization (OCBO)	Behaviors that benefit the organization in general (e.g., adheres to informal rules devised to maintain order; Williams & Anderson, 1991), which include civic virtue, sportsmanship, loyalty, extra-role behaviors toward the organization, and traditional OCBO (often measured as a composite of these dimensions).
Citizenship Behavior targeting the Supervisor (OCBS)	Behaviors that benefit the supervisor in general (e.g., assists supervisor with his/her work when not asked; Williams & Anderson, 1991)
<i>CWB</i>	
CWB targeting the Organization	
(a) Organization-targeted Sabotage	Behaviors that "damage or disrupt the organization's operations by creating delays in production, damaging property, the destruction of relationships, ..." (Crino, 1994, p. 312)
(b) Organization-targeted Retaliation	"A subset of negative behaviors, those used to punish the organization and its representatives in response to perceived unfairness" (Skarlicki & Folger, 1997, p. 435)
(c) Employee Theft	Stealing things from the company (e.g., Greenberg, 1990)
(d) Organization-targeted Counterproductive Work Behavior	"Behaviors by an organizational member that results in harming the organization or its members" (Martinko, Gundlach, & Douglas, 2002, p. 37)
(e) Organizational Deviance	"Voluntary behavior that violates significant organizational norms and in so doing threatens the well-being of an organization, its members, or both" (Robinson & Bennett, 1995, p. 556).
CWB targeting the Supervisor	
(a) Supervisor-targeted Sabotage	Behaviors that "damage or disrupt the organization's operations by creating delays in production, damaging property, the destruction of relationships, ..." (Crino, 1994, p. 312)
(b) Supervisor-targeted Retaliation	"A subset of negative behaviors, those used to punish the organization and its representatives in response to perceived unfairness" (Skarlicki & Folger, 1997, p. 435)
(c) Supervisor-targeted Aggression	"Any behavior where the aggressor delivers a noxious stimulus to another person – with the intent of harming the other person – and expects that this noxious stimulus will harm the targeted victim" (Greenberg & Barling, 1999, p. 898)
<i>Support</i>	
Perceived Organizational Support (POS)	Employees' "global beliefs concerning the extent to which the organization values their contributions and cares about their well-being" (Eisenberger, Huntington, Hutchison, & Sowa, 1986, p. 501)
Perceived Supervisory Support (PSS)	Employees' "general views concerning the degree to which supervisors value their contributions and care about their well-being" (Eisenberger, Stinglhamber, Vandenberghe, Sucharski, & Rhoades, 2002, p. 565)
<i>Identification</i>	
Organizational Identification	"A perceived oneness with an organization and the experience of the organization's successes and failures as one's own" (Mael & Ashforth, 1992, p. 103)
Supervisory Identification	"Feeling of pride being a subordinate of the supervisor" (Chen, 2001, p. 651)
<i>Satisfaction</i>	
Work (Job) Satisfaction	"The pleasurable emotional state resulting from the appraisal of one's job as achieving or facilitating the achievement of one's job values" (Locke, 1969, p. 316)
Supervisory Satisfaction	The positive emotional state resulting from the appraisal of one's supervisor
<i>Commitment</i>	
Organizational Commitment	
Affective Commitment	"The attachment of an individual's fund of affectivity and emotion" to their organization (Allen & Meyer, 1990; Kanter, 1968, p. 507)
Continuance Commitment	Individuals' perceived connectedness with their employer due to a "profit associated with continued participation and a "cost" associated with leaving" (Allen & Meyer, 1990; Kanter, 1968, p. 504)
Normative Commitment	Individuals' perceived connectedness with their employer due to a belief that staying "is the "right" and moral thing to do" (Allen & Meyer, 1990; Wiener, 1982, p.421)
Supervisory Commitment	The attachment of an individual's fund of affectivity and emotion to their supervisor (e.g., Byrne, 1999)
<i>Trust</i>	
Supervisory Trust	"The willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party" (Mayer, Davis, & Schoorman, 1995, p. 712); Employees' trust can be directed toward their employer and/or their supervisor
Organizational Trust	
<i>Job Performance</i>	
	"The aggregated value to the organization of the discrete behavioral episodes that an individual performs over a standard interval of time" (Motowidlo, Borman, & Schmit, 1997, p. 71)
<i>Leader-member Exchange (LMX)</i>	
	The quality of the dyadic relationship between subordinate and their supervisor (Dansereau, Graen, & Haga, 1975)



Note. Number on top of each bar signifies the (k) number of studies from the meta-analytic database coded as measuring the signified type and source. Under this number in parentheses is the (N) total sample size across all studies coded as measuring the type/source combination. This figure represents all published studies meeting our coding criteria, up until the year 2012. Note that not all types and sources of justice are included in every study, yet some studies contain multiple types and/or sources.

Fig. 1. Sources specified in justice measures used in empirical justice research ($k = 647$, $N = 235,682$).

Coding of outcome variables

In order to capture the relationship between justice perceptions and various target-specific outcomes (e.g., organizational commitment, supervisory trust), we also coded for the target to whom these outcomes were directed. For example, one of the variables studied in Lee and Farh (1999) study of gender and justice perceptions included Roberts and O'Reilly (1974) measure of supervisory trust. The items on this measure take the format: "How free do you feel to discuss with your immediate supervisor the problems and difficulties you have in your job, without jeopardizing your position or having it held against you?" Measures with these sorts of items were coded as target = supervisor. We followed the same procedures described above for coding the target of outcomes, which involved scanning hypotheses and method sections for evidence that the measure cited had been modified or not, and/or, when needed, obtaining the measures from original source materials, the scale developers themselves, or the study authors.

Job performance was coded as a supervisor-directed variable, as it has been consistently treated this way in past research, spanning the areas of justice (Cropanzano et al., 2002; Masterson et al., 2000; Rupp & Cropanzano, 2002), leader member exchange (Settoon et al., 1996; Wayne, Shore, & Liden, 1997), and organizational commitment (Becker, Billings, Eveleth, & Gilbert, 1996). As supervisors are more proximately impacted by employee performance, and are commonly the most centrally involved party in performance appraisal contexts, research consistently shows the effects of supervisor-focused perceptions and attitudes to more strongly predict performance, as compared to organization-focused perceptions and attitudes (e.g., Aryee, Budhwar, & Chen, 2002; Becker & Kernan, 2003; Bentein, Stinglhamber, & Vandenberghe, 2002; Dirks & Ferrin, 2002).

In addition, for parsimony, due to large number of overlapping variables and subfacets used to measure OCBO, counterproductive behaviors targeting the organization, and counterproductive behaviors targeting the supervisor, variable clusters were formed for these three variables. Specifically, OCBO included civic virtue, sportsmanship, loyalty, organization-directed extra-role behavior, and traditional OCBO (often measured as a composite of these dimensions). CWB toward organization included organization-directed sabotage, retaliation, theft, deviance, and CWB. CWB toward supervisor included supervisor-directed sabotage, retaliation, and aggression.

Coding of inter-correlations

We then recorded the inter-correlations (i.e., zero-order correlation coefficients) among different types of justice (e.g., between distributive justice and procedural justice) directly from the studies' correlation matrices. If no correlation matrix was provided by the study authors, we calculated this information based on other statistical information available in the study (Lipsey & Wilson, 2001). Since we coded justice measures by both type and source, for those variables for which we could specify a source, we were able to record the zero-order correlation coefficients between different justice sources and types [e.g., between (coded) organizationally-focused justice and (coded) supervisory-focused justice or between (coded) organizationally-focused justice and procedural justice]. In addition, we recorded the correlations between types and sources of justice and the target-specific outcomes (e.g., between distributive justice and supervisory trust, between supervisory-focused justice and supervisory commitment). Lastly, we recorded the inter-correlations among all outcome variables (e.g., between supervisory trust and supervisory commitment). During the coding process, whenever multiple correlations were reported concerning the same bivariate relationship within an individual study (e.g., when the variables of interest were measured at multiple time points), we followed the protocol outlined in the meta-analysis literature for calculating a single "composite" correlation for that bivariate relationship, for that study (see Hunter & Schmidt, 2004; Lipsey & Wilson, 2001). Organizing the correlational data in this way allowed us to calculate the average effect sizes (i.e., averaged correlation coefficients) between different justice types, between different justice sources, between justice types and sources, between justice types and outcomes, between justice sources and outcomes, and between different outcomes. This then allowed us to construct multiple meta-analytic correlation matrices among variables with which to perform regressions and path analyses to test our hypotheses.

Reliability check

A team of research assistants carried out the coding. These assistants went through extensive coder training and practice so that they fully understood the information they needed to extract from the articles. Next, all of the coding was checked by two members of our research team (authors herein). These researchers read the original articles and examined if the coding was appropriate. When discrepancies arose, they were discussed and consensus

was reached on the proper coding. Lastly, a random sample of 150 articles were coded again by another member of the research team. This coding matched with the original coding 98% of the time. Again, any discrepancies were discussed and resolved via consensus.

Meta-analysis approach

We performed meta-analyses following the procedures developed by Hedges and colleagues (Borenstein, Hedges, Higgins, & Rothstein, 2009; Hedges & Olkin, 1985; Hedges & Vevea, 1998; Lipsey & Wilson, 2001). First, we performed the Fisher's z transformation ($z = .5 \ln \frac{1+r}{1-r}$) on each effect size (e.g., the zero-order correlation coefficients between justice types and outcomes). Second, we calculated the inverse variance weight (w) for each of the transformed effect sizes. In statistical terms, an effect size's inverse variance weight equals the sample size of that specific study minus three (i.e., $w = N - 3$), which is the inverse of the sampling variance (presented in this formula: $\sigma_z^2 = \frac{1}{N-3}$) of Fisher's z transformation of the Pearson product-moment correlation coefficients (r). Given that inverse variance weight is perfectly positively related to the sample size of each study, in meta-analysis, the effect sizes from larger samples have higher weights than the effect sizes from smaller samples in calculating the average effect size. This weighting approach allows us to correct the sampling error in effect size estimates in each individual empirical study in that larger samples are likely to have more accurate estimates of the population parameters as compared to smaller samples (Borenstein et al., 2009; Hedges & Olkin, 1985; Hedges & Vevea, 1998). Third, we used the SPSS Macro and Syntax for meta-analysis developed by Lipsey and Wilson (2001) to calculate the average effect sizes, standard errors, confidence intervals, and homogeneity tests results [Q -test, p value of Q test and random effects variance component (REVC)]; and to transfer the results back from Fisher's z to Pearson's r correlations. All of the meta-analyzed (averaged) correlation coefficients are based on random effects model results (Borenstein et al., 2009).

Results

As explained earlier, whereas a number of sources emerged as we coded the articles, such as coworkers and customers, these sources appeared far less frequently than supervisors and organizations. Indeed, we only found nine independent correlational studies that explicitly specified the source of (in)justice as coworkers or customers (e.g., Skarlicki et al., 2008). We believe that more studies are necessary for us to calculate meaningful averaged effect sizes between coworker-focused or customer-focused justice perceptions and outcomes targeting coworkers or customers, respectively. Thus, in the present study, we focused on the effect sizes between two major justice sources (i.e., supervisor and organization) and 18 outcomes targeting these two sources (see Table 1).

Combination of meta-analysis and regression

To test Hypotheses 1 and 2, we combined meta-analysis with hierarchical multiple regression techniques (cf., Colquitt et al., 2001) in order to compare the standardized regression coefficients associated with our constructs of interest, with all predictors (i.e., target similar justice, target dissimilar justice, and the three justice types) simultaneously present in the same regression equation. We then conducted follow-up regression analyses to make more specific comparisons relevant to our hypotheses [e.g., target similar justice source vs. justice types (H1); target similar vs. target dissimilar justice effects (H2)]. To do this, we first calculated the aver-

aged correlations among justice sources, justice types, and outcome variables, and used these averaged correlations to construct correlation matrices, on which regression analyses were performed.

Specifically, we calculated the meta-analytic (averaged) correlations between type-/source-based justice perceptions and each of the 18 target-specific outcomes (i.e., outcomes targeting the supervisor or the organization). Table 2 provides the meta-analytic correlations between the justice variables and the nine outcomes targeting the organization, along with their corresponding 95% confidence intervals and homogeneity test results. Table 3 provides the same information for the nine outcomes targeting the supervisor. Next, we calculated the meta-analytic (i.e., averaged) intercorrelations between type- and source-based justice perceptions. Table 4 provides these estimates, along with their corresponding 95% confidence intervals and homogeneity test results.

Based on the averaged correlations among all of the variables presented in Tables 2–4, we constructed 18 meta-analytic correlation matrices⁵ (corresponding to each outcome variable), and computed the harmonic mean of the sample sizes for each of the 18 correlation matrices. The calculation formula is [harmonic mean = $j / (1/N_1 + 1/N_2 + \dots + 1/N_j)$], where j refers to the number of averaged correlations in the matrix and N refers to the sample size for each averaged correlation in the matrix (cf. Colquitt et al., 2001; Viswesvaran & Ones, 1995). Then, based on the 18 correlation matrices and their harmonically averaged sample sizes, we performed hierarchical multiple regression analyses with justice variables (both justice sources and justice types) serving as predictors and each of the 18 target-specific outcomes serving as the criterion variable in each regression analysis (cf. Colquitt et al., 2001; Viswesvaran & Ones, 1995).

For each regression analysis (corresponding to each target-specific outcome variable) we followed the following procedure: In Step 1 we entered the justice types simultaneously (i.e., DJ, PJ, and IJ; see Step 1a in Tables 5 and 6). In Step 2 we entered both matched and mismatched justice sources (see Step 2a in Tables 5 and 6). This tells us the variance accounted for in each outcome by multifoci justice variables (both target similar and dissimilar), over and above the effects of justice types. Then, we ran additional regression analyses to further explore our hypotheses. First, to compare the incremental validity of justice sources over types to the incremental validity of justice types over sources, a second regression analysis was carried out, which is reported in Steps 1b and 2b. In contrast to Steps 1a and 2a, Steps 1b and 2b show the additional variance in outcomes accounted for by justice types, above and beyond the effect of justice sources. Second, to compare the relative predictive validity of target similar justice variables to target dissimilar justice variables, we ran two more regressions, with the justice types entered in the first step (Step 1a) and either the addition of target dissimilar justice (Step 2c), or target similar justice (Step 2d), as the second step. We walk through all of these analyses below. Table 5 presents all these regression analyses conducted on the nine outcomes targeting the organization. Table 6 presents all these regression analyses conducted on the nine outcomes targeting the supervisor.

H1: Does target-matched, multifoci justice more strongly predict outcomes, compared to the justice types?

Hypothesis 1 stated that multifoci justice perceptions would more strongly predict target similar outcomes as compared to type-based justice perceptions (i.e., distributive, procedural, and interactional justice perceptions). As is shown in Table 5 (the nine

⁵ These 18 matrices can be reconstructed from the averaged correlations reported in Tables 2–4, and are available from the authors upon request.

Table 2

Meta-analytic correlations between justice (separated by source and type) and outcomes targeting the organization.

Justice	k	N	$\bar{r}$	95% CI	Q	p	REVC	k	N	$\bar{r}$	95% CI	Q	p	REVC
Perceived organizational support								Organizational trust						
SJ	22	8675	.49	(.44, .54)	321.42	.00	.03	23	5257	.45	(.39, .49)	106.17	.00	.02
OJ	25	6987	.52	(.46, .58)	227.77	.00	.04	36	6792	.53	(.48, .57)	335.34	.00	.04
DJ	18	8303	.47	(.40, .55)	148.35	.00	.04	24	4379	.39	(.30, .47)	236.77	.00	.05
PJ	32	13,650	.50	(.45, .55)	360.72	.00	.03	35	8781	.47	(.43, .52)	136.27	.00	.02
IJ	18	7413	.42	(.37, .46)	152.81	.00	.01	23	4349	.38	(.29, .47)	283.55	.00	.06
Organizational identification								OCBO						
SJ	2	667	.30	(.15, .43)	3.73	.06	.01	42	11,508	.23	(.20, .26)	211.99	.00	.01
OJ	16	4311	.38	(.31, .44)	56.09	.00	.02	41	11,008	.20	(.16, .23)	180.89	.00	.01
DJ	10	2755	.34	(.24, .43)	37.52	.00	.02	35	16,070	.20	(.17, .26)	328.74	.00	.02
PJ	9	2421	.34	(.24, .42)	14.21	.01	.01	62	24,278	.23	(.20, .28)	1163.02	.00	.04
IJ	7	1515	.28	(.21, .35)	4.47	.34	.00	35	9960	.20	(.17, .24)	283.43	.00	.01
CWB – organization								Affective commitment – org						
SJ	24	5226	–.17	(–.21, –.13)	166.09	.00	.02	26	7294	.32	(.26, .38)	124.09	.00	.02
OJ	27	13,705	–.22	(–.26, –.18)	288.82	.00	.01	58	22,412	.40	(.36, .43)	646.87	.00	.02
DJ	17	5102	–.15	(–.21, –.08)	84.32	.00	.02	37	10,297	.34	(.30, .39)	98.05	.00	.01
PJ	24	13,043	–.18	(–.27, –.08)	544.47	.00	.06	63	24,438	.40	(.36, .45)	451.57	.00	.03
IJ	21	11,027	–.15	(–.24, –.07)	555.51	.00	.05	40	15,860	.39	(.35, .44)	292.79	.00	.02
Continuance commitment – org								Normative commitment – org						
SJ	12	2672	–.03	(–.08, .03)	21.24	.03	.00	6	1109	.28	(.18, .37)	14.50	.01	.01
OJ	22	4896	–.02	(–.07, .03)	63.74	.00	.01	15	3187	.32	(.25, .38)	50.56	.00	.01
DJ	11	2845	.00	(–.09, .09)	55.07	.00	.02	10	2564	.26	(.16, .36)	51.03	.00	.02
PJ	14	3439	–.02	(–.09, .05)	47.08	.00	.01	12	3259	.30	(.26, .34)	16.57	.08	.00
IJ	18	3754	–.02	(–.06, .02)	21.89	.19	.00	11	2407	.28	(.22, .35)	28.65	.00	.01
Job satisfaction														
SJ	66	18,038	.41	(.38, .46)	857.50	.00	.04							
OJ	110	54,317	.42	(.40, .44)	3001.76	.00	.03							
DJ	98	51,656	.34	(.30, .38)	2511.13	.00	.05							
PJ	111	53,028	.39	(.35, .43)	3832.98	.00	.06							
IJ	61	18,559	.40	(.38, .44)	444.11	.00	.02							

Note. *k* = number of studies; *N* = number of participants; $\bar{r}$ = aggregated effect sizes; CI = confidence interval around aggregated effect sizes; *Q* = homogeneity test score; *p* = *p* value of the *Q* test result (a significant *Q* test result implies heterogeneity); REVC = random effects variance component (REVC = 0 means homogeneity); SJ = supervisory-focused justice; OJ = organizationally-focused justice; DJ = distributive justice; PJ = procedural justice; IJ = interactional justice; OCBO = organizational citizenship behaviors targeting the organization; CWB – organization refers to counterproductive behaviors toward the organization which include organization-directed sabotage, retaliation, theft, CWB, and organizational deviance. All the results are based on the random effects model (Lipsey & Wilson, 2001).

regression equations in Step 2a), with the exception of OCBO (and continuance commitment, which was not relevant to our hypothesis tests⁶), for the outcomes targeting the organization, organizationally-focused justice was a stronger predictor of organization-directed outcomes as compared to the three justice types, as is evidenced by the non-overlapping confidence intervals around the beta coefficients. As shown in Table 6 (the nine regression equations in Step 2a), across all nine outcome variables targeting the supervisor, supervisory-focused justice was a stronger predictor as compared to the justice types, as is evidenced by the non-overlapping confidence intervals around the beta coefficients. Taken together, these results provide support for Hypothesis 1.

To provide further support for Hypothesis 1, we also conducted usefulness analyses to compare the incremental predictive validity of multifoci justice perceptions above and beyond the justice type perceptions to the incremental predictive validity of the justice type perceptions above and beyond multifoci justice perceptions. As shown in Tables 5 and 6, the ΔR^2 from Step 1a to Step 2a represented the incremental variance explained by justice sources over and above the effects of justice types. We performed another set of hierarchical regression analyses on each of the justice outcomes to examine the comparative incremental variance explained by justice types over and above justice sources. Specifically, we entered the relevant multifoci justice

variables in Step 1 (see Step 1b in Tables 5 and 6) and entered the justice types in Step 2 (see Step 2b in Tables 5 and 6). We then compared the ΔR^2 from Step 1a to Step 2a to the ΔR^2 from Step 1b to Step 2b in Tables 5 and 6. We found that with only two exceptions (affective organizational commitment and supervisory trust), across the 18 justice outcomes, the incremental variance explained by justice sources over and above justice types is greater than it is for justice types over and above justice sources. This provides additional support for Hypothesis 1 from a relative incremental validity perspective, and supports the utility of modeling justice according to source (and in a way that subsumes the justice types). It should also be noted that the regression models tested not only included target similar justice, but also included target dissimilar justice, which seemed to also contribute to predicting additional variance in outcome variables. We will discuss this issue in more detail below.

H2: Does target similar justice predict outcomes better than target dissimilar outcomes?

Our second Hypothesis predicted that multifoci justice perceptions would more strongly predict outcomes directed at matched sources than mismatched (i.e., target dissimilar) sources. We also tested this prediction in two ways. First, we examined whether target similar effects were stronger than target dissimilar effects via a comparison of the resulting beta coefficients. As shown in Step 2a in Table 5, with the exception of OCBO, across the outcomes targeting the organization, organizationally-focused justice was a stronger predictor than supervisory-focused justice, as is evidenced by the non-overlapping confidence intervals around the beta coefficients. As shown in Step 2a in Table 6, with the exception of counterproductive behaviors toward the supervisor, across

⁶ Please note that continuance commitment was not a variable pertinent to any of our hypotheses, but was only included for completeness (i.e., to potentially aid future meta-analysts). Scales used to measure continuance commitment often suffer from unreliability. In addition, continuance commitment is less about one's current work conditions and more about external options (Allen & Meyer, 1990), and as such, justice researchers rarely make predictions regarding this variable.

Table 3

Meta-analytic correlations between justice (separated by source and type) and outcomes targeting the supervisor.

Justice	<i>k</i>	<i>N</i>	$\bar{r}$	95% CI	<i>Q</i>	<i>p</i>	REVC	<i>k</i>	<i>N</i>	$\bar{r}$	95% CI	<i>Q</i>	<i>p</i>	REVC
Perceived supervisory support								Supervisory trust						
SJ	12	3806	.64	(.57, .71)	192.50	.00	.05	36	10,349	.58	(.53, .62)	970.36	.00	.06
OJ	10	1837	.44	(.33, .54)	189.31	.00	.06	13	2511	.40	(.29, .50)	112.88	.00	.04
DJ	6	1520	.44	(.28, .65)	64.52	.00	.07	28	6474	.40	(.35, .45)	182.50	.00	.03
PJ	9	1640	.44	(.32, .53)	45.78	.00	.04	31	8127	.51	(.41, .60)	952.94	.00	.13
IJ	9	3186	.62	(.51, .73)	139.61	.00	.07	24	5370	.55	(.48, .61)	202.71	.00	.04
Supervisory identification								OCBS						
SJ	7	1397	.56	(.49, .62)	18.12	.01	.01	15	3683	.33	(.26, .39)	105.21	.00	.02
OJ	4	872	.29	(.23, .35)	3.13	.37	.00	19	5558	.17	(.08, .26)	140.81	.00	.03
DJ	4	786	.29	(.21, .40)	6.92	.07	.00	11	3073	.21	(.14, .28)	20.37	.01	.01
PJ	4	786	.37	(.31, .43)	3.30	.35	.00	17	4639	.21	(.14, .28)	58.94	.00	.01
IJ	6	1222	.50	(.37, .60)	39.83	.00	.04	16	3915	.30	(.22, .39)	143.17	.00	.04
CWB to supervisor								Supervisory commitment						
SJ	12	3259	-.27	(-.35, -.19)	64.10	.00	.02	11	2302	.60	(.53, .66)	56.05	.00	.02
OJ	11	3302	-.26	(-.35, -.20)	48.79	.00	.01	7	1671	.34	(.27, .40)	13.97	.03	.01
DJ	10	3141	-.16	(-.34, .03)	170.23	.00	.08	4	839	.38	(.29, .46)	6.21	.10	.00
PJ	5	1670	-.25	(-.32, -.20)	8.94	.03	.01	7	1407	.36	(.26, .45)	77.46	.00	.06
IJ	10	3211	-.25	(-.37, -.11)	80.23	.00	.04	12	2464	.45	(.27, .59)	280.45	.00	.12
Leader–member exchange								Job performance						
SJ	25	5595	.65	(.60, .70)	622.19	.00	.07	35	11,007	.27	(.19, .36)	1354.05	.00	.10
OJ	27	7685	.42	(.37, .47)	468.53	.00	.04	34	9678	.24	(.14, .34)	1356.71	.00	.11
DJ	30	8172	.37	(.30, .44)	491.93	.00	.06	32	7861	.16	(.11, .21)	86.88	.00	.01
PJ	30	8910	.43	(.38, .48)	610.01	.00	.06	49	11,909	.20	(.13, .26)	407.03	.00	.04
IJ	28	6608	.60	(.56, .66)	634.74	.00	.06	33	9803	.22	(.16, .29)	387.70	.00	.04
Supervisory satisfaction														
SJ	20	4825	.64	(.60, .67)	199.18	.00	.02							
OJ	17	4301	.41	(.32, .48)	192.00	.00	.04							
DJ	10	1904	.36	(.19, .51)	133.19	.00	.08							
PJ	16	3014	.40	(.30, .49)	251.49	.00	.05							
IJ	19	5463	.48	(.38, .56)	455.96	.00	.07							

Note. *k* = number of studies; *N* = number of participants; $\bar{r}$ = aggregated effect sizes; CI = confidence interval around aggregated effect sizes; *Q* = homogeneity test score; *p* = *p* value of the *Q* test result (a significant *Q* test result implies heterogeneity); REVC = random effects variance component (REVC = 0 means homogeneity); SJ = supervisory-focused justice; OJ = organizationally-focused justice; DJ = distributive justice; PJ = procedural justice; IJ = interactional justice; OCBS = organizational citizenship behaviors targeting the supervisor; CWB to supervisor refers to counterproductive behaviors to the supervisor which include supervisor-directed sabotage, retaliation, and aggression. All the results are based on the random effects model (Lipsey & Wilson, 2001).

Table 4

Meta-analytic correlation matrix among justice types/sources.

Variables	1		2		3		4	
	$\bar{r}$ (95% CI)	<i>Q</i> (<i>p</i>)	$\bar{r}$ (95% CI)	<i>Q</i> (<i>p</i>)	$\bar{r}$ (95% CI)	<i>Q</i> (<i>p</i>)	$\bar{r}$ (95% CI)	<i>Q</i> (<i>p</i>)
1. Supervisory-focused justice								
2. Organizationally-focused justice	.42 (.38, .45)	3163.75 (.00)						
	91 (24,171)	.07						
3. Distributive justice	.40 (.37, .43)	4170.25 (.00)	.43 (.39, .46)	5430.84 (.00)				
	131 (33,520)	.07	107 (44,485)	.07				
4. Procedural justice	.48 (.46, .51)	9898.45 (.00)	.43 (.40, .46)	5469.04 (.00)	.46 (.44, .49)	9594.46 (.00)		
	136 (74,044)	.08	108 (33,622)	.07	261 (97,967)	.08		
5. Interactional justice	.53 (.49, .57)	5749.75 (.00)	.43 (.40, .46)	4427.39 (.00)	.42 (.39, .44)	2808.57 (.00)	.53 (.50, .55)	8954.78 (.00)
	63 (22,016)	.12	94 (31,618)	.07	153 (42,429)	.05	189 (95,608)	.07

Note. *k* = number of studies; *N* = number of participants; $\bar{r}$ = aggregated correlation coefficients; CI = confidence interval around aggregated correlation coefficients; *Q* = homogeneity test score; *p* = *p* value of the *Q* test result (a significant *Q* test result implies heterogeneity); REVC = random effects variance component (REVC = 0 means homogeneity); all the results are based on the random effects model (Lipsey & Wilson, 2001).

the outcomes targeting the supervisor, supervisory-focused justice was a stronger predictor than organizationally-focused justice, also evidenced by the non-overlapping confidence intervals around the beta coefficients. This provides support for Hypothesis 2.

One potential limitation with testing Hypothesis 2 in this way is that the analysis does not provide information about the utility gained by moving from target dissimilar to target similar predictions (while still accounting for the justice types in the model). Thus, as mentioned above, we performed additional regression analyses to test Hypothesis 2 more directly. Specifically, for crite-

ria variables targeting the organization, we first entered the three justice types in the regression equation (see Step 1a in Table 5), followed by supervisory-focused justice (representing target dissimilar effects; see Step 2c in Table 5). For comparison, we ran alternative regression analyses, where organizationally-focused justice was instead entered in this second step (representing target similar effects; see Step 2d in Table 5). We performed the same analyses for outcomes targeting the supervisor (see Step 2c and Step 2d in Table 6). Thus, the ΔR^2 from Step 1a to Step 2c represents the incremental variance explained by adding the target dis-

Table 5
Meta-analytic hierarchical multiple regression analysis results – justice predicting outcomes targeting the organization.

Justice	β (95%CI) Step 1a	β (95%CI) Step 2a	β (95%CI) Step 1b	β (95%CI) Step 2b	β (95%CI) Step 2c	β (95%CI) Step 2d
<i>Perceived organizational support</i>						
DJ	.27 (.26, .28)	.18 (.16, .19)		.18 (.16, .19)	.23 (.22, .25)	.20 (.19, .21)
PJ	.29 (.28, .31)	.19 (.18, .21)		.19 (.18, .21)	.24 (.22, .25)	.23 (.22, .25)
IJ	.15 (.14, .17)	.02 (.01, .04)		.02 (.01, .04)	.07 (.05, .08)	.09 (.07, .10)
SJ		.20 (.19, .22)	.33 (.32, .34)	.20 (.19, .22)	.25 (.23, .26)	
OJ		.27 (.25, .28)	.38 (.37, .40)	.27 (.25, .28)		.30 (.28, .31)
N	17,747					
R ²	.34	.43	.36	.43	.38	.40
ΔR^2		.09***		.07***	.04***	.06***
<i>Organizational trust</i>						
DJ	.19 (.17, .21)	.08 (.07, .10)		.08 (.07, .10)	.15 (.14, .17)	.10 (.09, .12)
PJ	.31 (.29, .33)	.20 (.18, .22)		.20 (.18, .22)	.26 (.24, .27)	.24 (.22, .25)
IJ	.14 (.12, .16)	.00 (–.02, .02)		.00 (–.02, .02)	.06 (.04, .07)	.06 (.04, .08)
SJ		.18 (.16, .20)	.28 (.26, .29)	.18 (.16, .20)	.24 (.22, .25)	
OJ		.33 (.32, .35)	.41 (.40, .43)	.33 (.32, .35)		.36 (.34, .38)
N	12,844					
R ²	.27	.38	.34	.38	.31	.36
ΔR^2		.11***		.04***	.04***	.09***
<i>Organizational identification</i>						
DJ	.21 (.18, .24)	.15 (.12, .18)		.15 (.12, .18)	.20 (.16, .23)	.16 (.13, .19)
PJ	.19 (.16, .23)	.13 (.10, .17)		.13 (.10, .17)	.17 (.14, .20)	.15 (.11, .18)
IJ	.09 (.06, .12)	.01 (–.02, .05)		.01 (–.02, .05)	.05 (.02, .08)	.04 (.01, .07)
SJ		.08 (.05, .11)	.17 (.14, .20)	.08 (.05, .11)	.12 (.08, .15)	
OJ		.22 (.19, .25)	.31 (.28, .34)	.22 (.19, .25)		.23 (.20, .27)
N	4379					
R ²	.16	.21	.17	.21	.17	.20
ΔR^2		.05***		.04***	.01***	.04***
<i>OCBO</i>						
DJ	.10 (.09, .11)	.07 (.06, .08)		.07 (.06, .08)	.08 (.07, .09)	.08 (.07, .09)
PJ	.14 (.13, .15)	.10 (.09, .11)		.10 (.09, .11)	.11 (.10, .12)	.12 (.11, .13)
IJ	.09 (.07, .10)	.03 (.02, .05)		.03 (.02, .05)	.04 (.03, .06)	.07 (.05, .08)
SJ		.11 (.10, .12)	.18 (.17, .19)	.11 (.10, .12)	.12 (.11, .13)	
OJ		.07 (.06, .08)	.13 (.12, .14)	.07 (.06, .08)		.09 (.07, .10)
N	23,446					
R ²	.07	.09	.08	.09	.08	.08
ΔR^2		.02***		.01***	.01***	.01***
<i>CWB to organization</i>						
DJ	–.07 (–.09, –.06)	–.03 (–.04, –.01)		–.03 (–.04, –.01)	–.06 (–.07, –.04)	–.03 (–.05, –.02)
PJ	–.12 (–.13, –.10)	–.07 (–.09, –.06)		–.07 (–.09, –.06)	–.10 (–.11, –.08)	–.08 (–.10, –.07)
IJ	–.06 (–.08, –.04)	–.01 (–.02, .01)		–.01 (–.02, .01)	–.03 (–.05, –.01)	–.02 (–.04, –.01)
SJ		–.06 (–.08, –.04)	–.10 (–.11, –.08)	–.06 (–.08, –.04)	–.08 (–.10, –.07)	
OJ		–.15 (–.17, –.14)	–.18 (–.19, –.18)	–.15 (–.17, –.14)		–.16 (–.18, –.15)
N	16,932					
R ²	.04	.06	.056	.061	.05	.06
ΔR^2		.02***		.005***	.01***	.02***
<i>Affective commitment – org</i>						
DJ	.15 (.14, .16)	.10 (.08, .11)		.10 (.08, .11)	.14 (.13, .15)	.10 (.09, .11)
PJ	.22 (.20, .23)	.17 (.16, .18)		.17 (.16, .18)	.20 (.19, .22)	.18 (.16, .19)
IJ	.21 (.20, .23)	.16 (.14, .17)		.16 (.14, .17)	.19 (.18, .21)	.17 (.15, .18)
SJ		.03 (.02, .04)	.19 (.17, .20)	.03 (.02, .04)	.06 (.05, .08)	
OJ		.21 (.20, .22)	.32 (.31, .34)	.21 (.20, .22)		.21 (.20, .22)
N	23,400					
R ²	.22	.25	.19	.25	.22	.25
ΔR^2		.03***		.06***	.00	.03***
<i>Continuance commitment⁶ – org</i>						
DJ	.00 (–.02, .03)	.01 (–.02, .03)		.01 (–.02, .03)	.01 (–.02, .03)	.01 (–.02, .03)
PJ	–.01 (–.04, .01)	–.01 (–.03, .02)		–.01 (–.03, .02)	–.01 (–.03, .02)	–.01 (–.04, .02)
IJ	–.01 (–.04, .01)	–.00 (–.03, .02)		–.00 (–.03, .02)	–.00 (–.03, .02)	–.01 (–.04, .01)
SJ		–.03 (–.05, .01)	–.03 (–.05, .00)	–.03 (–.05, .01)	–.03 (–.05, .00)	
OJ		–.01 (–.03, .02)	–.01 (–.03, .01)	–.01 (–.03, .02)		–.01 (–.04, .01)
N	8594					
R ²	.00	.00	.001	.00	.00	.00
ΔR^2		.00		.00	.00	.00
<i>Normative commitment – org</i>						
DJ	.12 (.10, .15)	.07 (.04, .10)		.07 (.04, .10)	.11 (.08, .13)	.08 (.05, .11)
PJ	.17 (.14, .20)	.11 (.08, .14)		.11 (.08, .14)	.14 (.11, .17)	.13 (.10, .16)
IJ	.14 (.11, .17)	.07 (.04, .10)		.07 (.04, .10)	.10 (.07, .13)	.10 (.07, .13)
SJ		.09 (.06, .12)	.18 (.15, .20)	.09 (.06, .12)	.12 (.09, .15)	
OJ		.18 (.15, .20)	.25 (.22, .27)	.18 (.15, .20)		.19 (.16, .22)
N	5800					

Table 5 (continued)

Justice	β (95%CI) Step 1a	β (95%CI) Step 2a	β (95%CI) Step 1b	β (95%CI) Step 2b	β (95%CI) Step 2c	β (95%CI) Step 2d
R^2	.12	.15	.13	.15	.13	.15
ΔR^2		.03***		.02***	.01***	.03***
<i>Job satisfaction</i>						
DJ	.15 (.14, .16)	.08 (.07, .08)		.08 (.07, .08)	.12 (.11, .13)	.09 (.08, .10)
PJ	.20 (.19, .21)	.12 (.11, .12)		.12 (.11, .12)	.15 (.14, .16)	.15 (.14, .16)
IJ	.23 (.22, .24)	.13 (.12, .14)		.13 (.12, .14)	.16 (.15, .17)	.18 (.17, .19)
SJ		.17 (.16, .18)	.28 (.27, .29)	.17 (.16, .18)	.20 (.19, .21)	
OJ		.21 (.21, .22)	.30 (.29, .31)	.21 (.21, .22)		.24 (.23, .25)
N	35,364					
R^2	.22	.28	.24	.28	.25	.26
ΔR^2		.06***		.04***	.03***	.04***

Note. β = standardized partial regression coefficients; CI = confidence interval around the standardized partial regression coefficients; N = number of participants; R^2 = Adjusted R^2 ; for comparison purposes, ΔR^2 reported under Step 2c and Step 2d all refers to the change of R^2 from Step 1a (i.e., from Step 1a to Step 2c; from Step 1a to Step 2d, respectively); SJ = supervisory-focused justice; OJ = organizationally-focused justice; DJ = distributive justice; PJ = procedural justice; IJ = interactional justice; OCBO = organizational citizenship behaviors targeting the organization; CWB to organization refers to counterproductive behaviors to the organization which include organization-directed sabotage, retaliation, theft, CWB, and organizational deviance. In order to check if multicollinearity biased our results, we examined the tolerance values and the variance inflation factor (VIF) of each regression equation. All the tolerance values are above .20 and all the VIF values are below 5, which suggest that multicollinearity is not a concern in our analyses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

Bold values indicate "target similarity effects".

similar justice effects beyond the justice types, and the ΔR^2 from Step 1a to Step 2d represents the incremental variance explained by adding the target similar justice effects above and beyond the justice types. Consistent with the findings described above, with the exception of OCBO and counterproductive work behavior toward supervisor, across these outcomes, we found that the former ΔR^2 s were smaller than the latter ΔR^2 s. Taken together, both approaches provide support for Hypothesis 2.

The role of cross-foci justice effects. As we mentioned at the start of our paper, although not directly hypothesized, we did, based on previous research (Lavelle et al., 2007, in press; Masterson et al., 2000; Rupp & Cropanzano, 2002) expect that employees' perceptions of justice about a source might impact their attitudes and behaviors toward sources other than the justice perception referent (though the magnitude of such target dissimilar effects was expected to be smaller as compared to target similar effects). Due to the centrality of one's supervisor, the most target dissimilar effects were expected for supervisory-focused justice impacting organization-directed outcomes. As reported above, our results substantiated this claim to some extent. First, we direct readers' attention to Step 2c in Tables 5 and 6. Although rarely as large as the beta coefficients for target similar justice effects (reported in Step 2d), the majority of the target dissimilar effects were statistically significant. Further, comparing the ΔR^2 values reported in Step 2d to those reported in Step 2a, we see for some variables, a slight increase in variance accounted for by adding both target similar and target dissimilar justice to the model (in comparison to only adding target similar justice). To explore if the addition of target dissimilar effects accounted for more incremental variance in organization-directed outcomes compared to supervisor-directed outcomes (hence suggesting a stronger role of supervisor cross-foci effects as compared to organization cross-foci effects), we computed and compared the average of the differences between Step2d and Step2a ΔR^2 s across organization-directed outcomes (.01) to that of supervisor-directed outcomes (.006). The result corroborated our expectation.

Combination of meta-analysis and path analysis to test for the mediating effect of social exchange (H3)

Our third Hypothesis predicted that the relationships between multifoci justice perceptions and target similar outcomes would be mediated by social exchange with the respective source; that such mediation effects would exist simultaneously for different

sources; and that these mediated target similar effects would exist above and beyond possible cross-foci effects. To test this prediction, we combined meta-analysis with path analysis techniques. First, we used the same procedures described above to calculate the matrix of meta-analytic correlations among organizationally-focused justice perceptions, supervisory-focused justice perceptions, the social exchange mediators targeting the organization (e.g., perceived organizational support), the social exchange mediators targeting the supervisor (e.g., leader-member exchange), OCBS, and OCBO. We chose to only include OCBS and OCBO in our structural model, both for parsimony (given our seven total outcome variables)⁷, and to more directly test Lavelle et al.'s (2007) target similarity model, which includes these two outcome variables only. Also OCBO and OCBS provide a single behavioral construct that has shown to manifest as behaviors directed both at the organization and the supervisor (Byrne, 2005; Karriker & Williams, 2009).

Table 7 presents this meta-analyzed correlation matrix. We excluded perceived supervisory support, supervisory identification, and supervisory commitment from this matrix because the inclusion of these three variables created empty cells in the matrix (cf. Hom, Caranikas-Walker, Prussia, & Griffeth, 1992). We then computed the harmonic mean of the sample sizes in this matrix (cf. Colquitt et al., 2001; Viswesvaran & Ones, 1995).

Next, based on the matrix and the averaged sample size, we constructed a theoretical mediation model containing the justice

⁷ We also conducted these same analyses using alternative outcome variables that were available within our meta-analytic database (e.g., CWBO, CWBS, job satisfaction, supervisory satisfaction, job performance). The results were consistent with those reported for OCBO and OCBS. That is, we found that social exchange with a particular source mediated the relationship between source-based justice perceptions and source-specific outcomes. It should be noted that the mediation effects we detected for CWBO and CWBS differed from Colquitt et al. (2013), who did not find social exchange to mediate the relationship between justice and CWB. In the discussion section, we explain how these two studies differ by virtue of how justice and outcomes were coded with respect to source, and how both the justice and CWB variables were coded and calculated differently across the two studies. Also, in their mediation model, Colquitt et al. (2013) used various social exchange variables as indicators of a second-order social exchange quality latent variable, and they did not differentiate the targets of outcomes in their mediation tests. In contrast, we included two justice sources, multiple source-specific social exchange variables, and target-specific outcomes in our mediation tests. As our approach is more closely aligned with what social exchange theory would suggest (i.e., reciprocation of socio-emotional benefits, targeted back at the specific exchange partner, Cropanzano & Mitchell, 2005), we feel our particular results represent a more robust test of the target similarity model.

Table 6
Meta-analytic hierarchical multiple regression analysis results – justice predicting outcomes targeting the supervisor.

Justice	β (95%CI) Step 1a	β (95%CI) Step 2a	β (95%CI) Step 1b	β (95%CI) Step 2b	β (95%CI) Step 2c	β (95%CI) Step 2d
<i>Perceived supervisory support</i>						
DJ	.19 (.17, .21)	.11 (.09, .13)		.11 (.09, .13)	.16 (.14, .18)	.13 (.11, .15)
PJ	.09 (.07, .11)	.02 (.01, .04)		.02 (.01, .04)	.06 (.04, .08)	.00 (–.02, .02)
IJ	.49 (.47, .51)	.33 (.31, .35)		.33 (.31, .35)	.46 (.44, .48)	.35 (.33, .37)
OJ		.10 (.08, .12)	.21 (.19, .23)	.10 (.08, .12)	.15 (.13, .17)	
SJ		.38 (.36, .40)	.55 (.53, .57)	.38 (.36, .40)		.40 (.38, .42)
N	5667					
R ²	.43	.54	.45	.54	.45	.53
ΔR^2		.11***		.09***	.02***	.10***
<i>Supervisory trust</i>						
DJ	.13 (.12, .15)	.07 (.05, .08)		.07 (.05, .08)	.10 (.09, .12)	.08 (.07, .10)
PJ	.26 (.25, .28)	.18 (.16, .19)		.18 (.16, .19)	.24 (.22, .26)	.19 (.17, .20)
IJ	.36 (.34, .37)	.23 (.22, .25)		.23 (.22, .25)	.33 (.32, .35)	.24 (.23, .26)
OJ		.06 (.05, .08)	.19 (.18, .21)	.06 (.05, .08)	.11 (.10, .13)	
SJ		.32 (.30, .33)	.50 (.49, .52)	.32 (.30, .33)		.33 (.31, .34)
N	12,327					
R ²	.38	.45	.38	.45	.39	.45
ΔR^2		.07***		.07***	.01***	.07***
<i>Supervisory identification</i>						
DJ	.06 (.03, .10)	.01 (–.03, .04)		.01 (–.03, .04)	.05 (.01, .09)	.00 (–.03, .04)
PJ	.13 (.09, .17)	.04 (–.01, .08)		.04 (–.01, .08)	.12 (.08, .16)	.04 (–.01, .07)
IJ	.41 (.37, .45)	.27 (.23, .31)		.27 (.23, .31)	.40 (.36, .44)	.27 (.23, .31)
OJ		–.01 (–.05, .03)	.07 (.04, .10)	–.01 (–.05, .03)	.05 (.01, .09)	
SJ		.40 (.36, .44)	.53 (.50, .57)	.40 (.36, .44)		.40 (.36, .44)
N	2735					
R ²	.27	.37	.32	.37	.27	.37
ΔR^2		.10***		.05***	.00	.10***
<i>OCBS</i>						
DJ	.09 (.07, .11)	.06 (.04, .08)		.06 (.04, .08)	.09 (.07, .11)	.06 (.04, .08)
PJ	.04 (.02, .06)	–.01 (–.03, .02)		–.01 (–.03, .02)	.04 (.02, .06)	–.01 (–.03, .01)
IJ	.24 (.22, .26)	.16 (.14, .19)		.16 (.14, .19)	.24 (.22, .26)	.16 (.14, .18)
OJ		–.02 (–.04, .01)	.04 (.02, .06)	–.02 (–.04, .01)	.02 (–.01, .04)	
SJ		.23 (.21, .25)	.31 (.29, .33)	.23 (.21, .25)		.23 (.21, .25)
N	9960					
R ²	.10	.13	.11	.13	.10	.13
ΔR^2		.03***		.02***	.00	.03***
<i>CWB to supervisor</i>						
DJ	–.02 (–.05, .01)	.03 (–.01, .06)		.03 (–.01, .06)	.02 (–.01, .05)	.00 (–.03, .03)
PJ	–.16 (–.19, –.13)	–.10 (–.12, –.07)		–.10 (–.12, –.07)	–.12 (–.15, –.09)	–.12 (–.15, –.09)
IJ	–.16 (–.19, –.13)	–.08 (–.11, –.05)		–.08 (–.11, –.05)	–.12 (–.15, –.10)	–.10 (–.13, –.07)
OJ		–.14 (–.17, –.11)	–.18 (–.20, –.15)	–.14 (–.17, –.11)	–.16 (–.19, –.13)	
SJ		–.14 (–.16, –.11)	–.20 (–.22, –.17)	–.14 (–.16, –.11)		–.16 (–.19, –.13)
N	7155					
R ²	.08	.11	.10	.11	.10	.10
ΔR^2		.03***		.01***	.02***	.02***
<i>Supervisory commitment</i>						
DJ	.20 (.17, .23)	.12 (.10, .15)		.12 (.10, .15)	.18 (.15, .21)	.13 (.11, .16)
PJ	.10 (.07, .13)	–.01 (–.04, .02)		–.01 (–.04, .02)	.08 (.05, .11)	–.01 (–.04, .02)
IJ	.31 (.28, .34)	.15 (.12, .18)		.15 (.12, .18)	.29 (.26, .32)	.15 (.12, .18)
OJ		.04 (.01, .07)	.11 (.08, .14)	.04 (.01, .07)	.11 (.08, .14)	
SJ		.47 (.44, .49)	.56 (.53, .58)	.47 (.44, .49)		.47 (.44, .50)
N	4167					
R ²	.25	.40	.37	.40	.26	.40
ΔR^2		.14***		.03***	.01***	.14***
<i>Leader–member exchange</i>						
DJ	.11 (.10, .12)	.02 (.01, .03)		.02 (.01, .03)	.07 (.06, .08)	.04 (.03, .05)
PJ	.12 (.11, .13)	.01 (–.01, .02)		.01 (–.01, .02)	.09 (.08, .10)	.02 (.01, .03)
IJ	.49 (.48, .50)	.32 (.31, .33)		.32 (.31, .33)	.46 (.44, .47)	.34 (.32, .35)
OJ		.09 (.08, .10)	.18 (.17, .19)	.09 (.08, .10)	.16 (.14, .17)	
SJ		.43 (.42, .44)	.58 (.56, .59)	.43 (.42, .44)		.45 (.43, .46)
N	15,740					
R ²	.39	.52	.45	.52	.40	.51
ΔR^2		.13***		.07***	.02***	.12***
<i>Job performance</i>						
DJ	.05 (.04, .07)	–.00 (–.02, .01)		–.00 (–.02, .01)	.02 (.01, .03)	.03 (.01, .04)
PJ	.10 (.08, .11)	.03 (.02, .05)		.03 (.02, .05)	.07 (.05, .08)	.06 (.04, .07)
IJ	.15 (.13, .16)	.06 (.04, .07)		.06 (.04, .07)	.11 (.10, .13)	.08 (.06, .10)
OJ		.13 (.12, .14)	.15 (.14, .17)	.13 (.12, .14)	.16 (.14, .17)	
SJ		.17 (.16, .19)	.21 (.19, .22)	.17 (.16, .19)		.19 (.18, .21)
N	19,593					

Table 6 (continued)

Justice	β (95%CI) Step 1a	β (95%CI) Step 2a	β (95%CI) Step 1b	β (95%CI) Step 2b	β (95%CI) Step 2c	β (95%CI) Step 2d
R^2	.06	.10	.09	.10	.078	.084
ΔR^2		.04***		.01***	.018***	.024***
<i>Supervisory satisfaction</i>						
DJ	.15 (.13, .17)	.05 (.03, .06)		.05 (.03, .06)	.10 (.08, .12)	.07 (.06, .09)
PJ	.15 (.13, .17)	.02 (.00, .04)		.02 (.00, .04)	.11 (.10, .13)	.04 (.02, .06)
IJ	.34 (.32, .36)	.14 (.12, .16)		.14 (.12, .16)	.30 (.28, .32)	.16 (.14, .18)
OJ		.12 (.10, .14)	.17 (.16, .19)	.12 (.10, .14)	.19 (.17, .21)	
SJ		.49 (.47, .51)	.57 (.55, .58)	.49 (.47, .51)		.51 (.49, .53)
N	8631					
R^2	.28	.45	.43	.45	.30	.44
ΔR^2		.18***		.02***	.03***	.17***

Note. β = standardized partial regression coefficients; CI = confidence interval around the standardized partial regression coefficients; N = number of participants; R^2 = Adjusted R^2 ; for comparison purposes, ΔR^2 reported under Step 2c and Step 2d all refers to the change of R^2 from Step 1a (i.e., from Step 1a to Step 2c; from Step 1a to Step 2d, respectively); SJ = supervisory-focused justice; OJ = organizationally-focused justice; DJ = distributive justice; PJ = procedural justice; IJ = interactional justice; OCBS = organizational citizenship behaviors targeting the supervisor; CWB to supervisor refers to counterproductive behaviors to the supervisor which include supervisor-directed sabotage, retaliation, and aggression. In order to check if multicollinearity biased our results, we examined the tolerance values and the variance inflation factor (VIF) of each regression equation. All the tolerance values are above .20 and all the VIF values are below 5, which suggest that multicollinearity is not a concern in our analyses.

*** $p < .001$; ** $p < .01$; * $p < .05$.

Bold values indicate “target similarity effects”.

sources, social exchange mediators, OCBO and OCBS with paths reflecting the target similar effects only [i.e., no cross-foci effects (paths) were allowed in the model]. In addition to paths reflecting the indirect effects of justice sources on OCB targeting the matched sources, we also included the direct effect of organizationally-focused justice on OCBO and the direct effect of supervisory-focused justice on OCBS. However, both direct paths in a model containing the social exchange mediators were not significant and their addition significantly decreased the overall model fit. This indicates that when social exchange mediators are included in the model (i.e., indirect effects of justice sources on OCB toward the matched sources are modeled), the direct effects of justice sources on OCB toward the matched sources are substantially attenuated, which is consistent with our expectation (suggesting that multifoci justice phenomena are beyond what a bandwidth-fidelity explanation would suggest). Thus, we removed these two direct effect paths from the model and found the resulting mediation model demonstrated good fit with the data ($\chi^2 = 469.04$, $df = 24$, $CFI = .90$, $NFI = .90$, $GFI = .92$, $RMSEA = .09$).

Although this represents a target similarity model in its “purest” form, we recognize this is not a direct test of our proposed Hypothesis 3, in that we also acknowledged the likelihood of cross-foci effects. That is, although target similarity effects are expected to be more robust than target dissimilar effects, target dissimilar effects are not completely unexpected, and have been found on occasion in the empirical multifoci research (e.g., Masterson et al., 2000; Olkkonen & Lipponen, 2006; Rupp & Cropanzano, 2002; Stinglhamber & De Cremer, 2008). Thus, we carried out supplementary analyses to explore the potential for target dissimilar effects. Including all possible cross-foci paths significantly decreased the fit of our overall model, likely resulting from both the imprecise modeling of the phenomena, as well as the loss of degrees of freedom with which to fit the model due to the addition of 22 cross-foci paths. Thus, to be as conservative as possible, we modeled only those target dissimilar paths that modification indices revealed would increase the fit of the model. Of the 22 potential cross-foci paths, only the four target dissimilar paths from supervisory-focused justice perceptions to organization-directed social exchange (i.e., POS and organizational trust), and from organizationally-focused justice perceptions to supervisor-directed social exchange (i.e., LMX and supervisory trust) had any marked impact on model fit (although the increase in model fit as a result of adding these four paths was not substantial; see Fig. 2, where

the target dissimilar paths are demarcated by dashed lines). Further, as is shown in Fig. 2, the path coefficients of target dissimilar paths are of weaker magnitude compared to their target similar counterparts.

We then conducted path analyses, using recommended bootstrapping techniques (Hayes, 2009; Preacher & Hayes, 2004, 2008), to test for the mediating effects specified in Hypothesis 3. We used the statistical software AMOS 19 to perform all the model fit tests and mediation analyses. These analyses revealed that the total indirect effect of organizationally-focused justice perceptions on OCBO via the joint mediation effects of organizational identification, organizational affective commitment, perceived organizational support, and organizational trust was significant (total indirect effect = .21, 95% bootstrapping bias-corrected CI = (.18, .25), $p = .001$). In addition, the total indirect effect of supervisory-focused justice perceptions on OCBS via the joint mediation effects of leader-member exchange and supervisory trust was significant (total indirect effect = .28, 95% bootstrapping bias-corrected CI = (.24, .31), $p = .001$).

Our results support our predictions that social exchange explains the association between perceptions of justice about a source and outcomes directed at that source. As further support for our arguments, our analyses revealed that the total indirect effect of organizationally-focused justice perceptions on OCBS via the joint mediation effects of leader-member exchange and supervisory trust (total indirect effect = .10, 95% bootstrapping bias-corrected CI = (.08, .12), $p = .01$) and the total indirect effect of supervisory-focused justice perceptions on OCBO via the joint mediation effects of perceived organizational support and organizational trust (total indirect effect = .09, 95% bootstrapping bias-corrected CI = (.06, .11), $p = .01$) were both significantly weaker than the total indirect effects of matched justice sources on OCB targeting the matched sources, which is evidenced by the non-overlapping confidence intervals of these total indirect effects. For completeness, Table 8 presents all the single indirect effects included in the model. Although in the present study, we did not intend to compare the relative strength of these single indirect effects, we hope this Table can be useful for justice researchers who are interested in comparing the relative mediating role of the various social exchange variables in the relationships between justice sources and OCB targeting matched and/or mismatched sources. As one last supplementary set of analyses, we compared the meta-analytic mediation effects outlined in Hypothesis 3 (i.e.,

Table 7

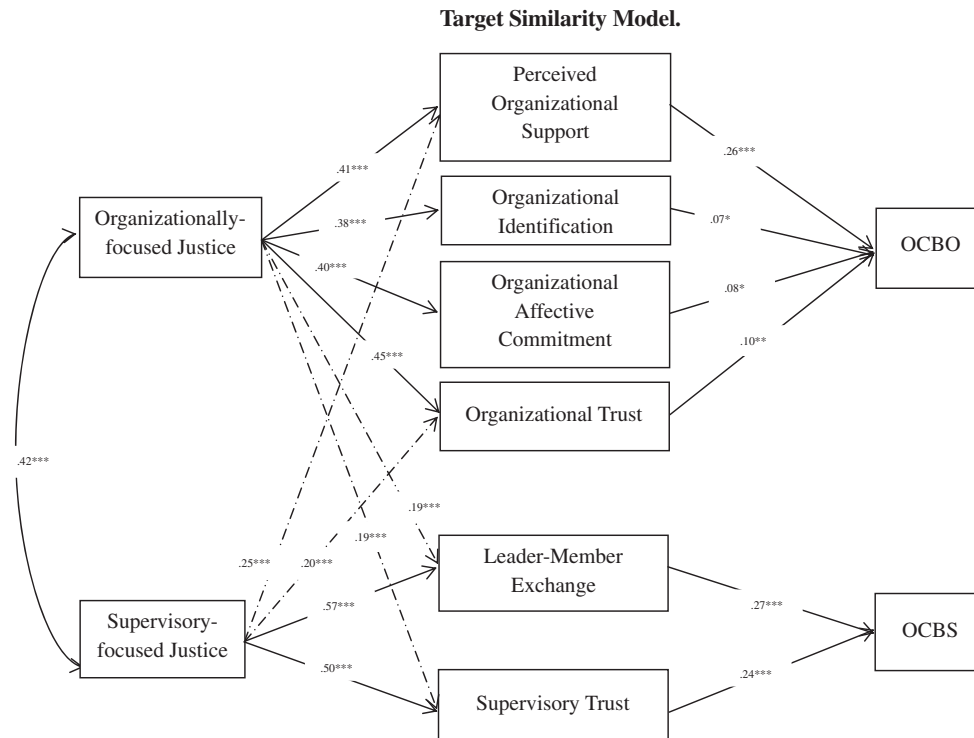
Meta-analytic correlation matrix among organizationally-focused justice, supervisory-focused justice, social exchange with organization and supervisor, and organizational citizenship behaviors targeting the organization and supervisor.

Variables	1		2		3		4		5	
	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)
	k (N)	REVC	k (N)	REVC	k (N)	REVC	k (N)	REVC	k (N)	REVC
1. OJ										
2. SJ	.42 (.38, .45)	3163.75 (.00)								
	91 (24,171)	.07								
3. POS	.52 (.46, .58)	227.77 (.00)	.49 (.44, .54)	321.42 (.00)						
	25 (6987)	.04	22 (8675)	.03						
4. OT	.53 (.48, .57)	335.34 (.00)	.45 (.39, .49)	106.17 (.00)	.55 (.36, .70)	123.98 (.00)				
	36 (6792)	.04	23 (5257)	.02	8 (2546)	.11				
5. AC	.40 (.36, .43)	646.87 (.00)	.32 (.26, .38)	124.09 (.00)	.56 (.48, .63)	408.57 (.00)	.54 (.46, .61)	290.83 (.00)		
	58 (22,412)	.02	26 (7294)	.02	23 (6763)	.08	24 (5,759)	.06		
6. OI	.38 (.31, .44)	56.09 (.00)	.30 (.15, .43)	3.73 (.06)	.48 (–.06, .80)	31.04 (.00)	.52 ^a	–	.65 (.31, .84)	18.71 (.00)
	16 (4311)	.02	2 (667)	.01	2 (375)	.17	1 (216)	–	2 (370)	.10
7. LMX	.42 (.37, .47)	468.53 (.00)	.65 (.60, .70)	622.19 (.00)	.28 (.22, .33)	31.09 (.00)	.24 (.15, .32)	12.14 (.02)	.26 (.21, .31)	41.53 (.00)
	27 (7685)	.04	25 (5595)	.07	10 (3287)	.01	5 (1031)	.01	14 (3370)	.01
8. ST	.40 (.29, .50)	112.88 (.00)	.58 (.53, .62)	970.36 (.00)	.27 (.07, .44)	24.92 (.00)	.28 (.19, .54)	26.70 (.00)	.23 (.13, .32)	201.95 (.00)
	13 (2511)	.04	36 (10,349)	.06	4 (963)	.04	8 (1654)	.04	18 (3445)	.06
9. OCBO	.20 (.16, .23)	180.89 (.00)	.23 (.20, .26)	211.99 (.00)	.35 (.20, .48)	176.55 (.00)	.28 (.12, .43)	155.26 (.00)	.35 (.26, .44)	778.44 (.00)
	41 (11,008)	.01	42 (11,508)	.01	12 (3931)	.07	11 (3758)	.07	35 (12,231)	.08
10. OCBS	.17 (.08, .26)	140.81 (.00)	.33 (.26, .39)	105.21 (.00)	.12 (.05, .18)	.60 (.44)	.12 (.03, .21)	.00 (1.00)	.22 (.12, .31)	9.42 (.02)
	19 (5558)	.03	15 (3683)	.02	2 (868)	.00	2 (433)	.00	4 (1363)	.01
Variables	6		7		8		9			
	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)	$\bar{r}$ (95%CI)	Q (p)		
	k (N)	REVC	k (N)	REVC	k (N)	REVC	k (N)	REVC		
1. OJ										
2. SJ										
3. POS										
4. OT										
5. AC										
6. OI										
7. LMX	.25 ^a	–								
	1 (216)	–								
8. ST	.14 ^a	–	.46 (.19, .67)	25.00 (.00)						
	1 (263)	–	3 (538)	.07						
9. OCBO	.33 (–.11, .63)	305.89 (.00)	.33 (.24, .40)	103.53 (.00)	.20 (.07, .32)	246.84 (.00)				
	5 (3091)	.19	14 (3851)	.03	12 (3535)	.07				
10. OCBS	.18	–	.38 (.27, .47)	55.79 (.00)	.38 (.16, .57)	31.78 (.00)	.35 (.26, .44)	158.46 (.00)		
	1 (216)	–	7 (2391)	.02	4 (809)	.05	15 (3775)	.04		

Note. k = number of studies; N = number of participants; $\bar{r}$ = aggregated correlation coefficients; CI = confidence interval around aggregated correlation coefficients; Q = homogeneity test score; p = p value of the Q test result (a significant Q test result implies heterogeneity); REVC = random effects variance component (REVC = 0 means homogeneity); OJ = organizationally-focused justice; POS = perceived organizational support; OT = organizational trust; AC = organizational affective commitment; OI = organizational identification; OCBO = organizational citizenship behaviors targeting the organization; SJ = supervisory-focused justice; LMX = leader-member exchange; ST = supervisory trust; OCBS = organizational citizenship behaviors targeting the supervisor.

We excluded supervisory identification, supervisory commitment, and perceived supervisory support from this matrix because the inclusion of these three variables created empty cells in the matrix.

^a Only one individual empirical study reported the correlation coefficient, thus, we cannot calculate the aggregated correlation coefficient and its confidence interval. We used the harmonic mean of the sample sizes ($[45 / (1/N_1 + 1/N_2 + \dots + 1/N_{45})] = 1106$ in path analyses ($j = 45$). All the results are based on the random effects model (Lipsey & Wilson, 2001).



Note. *** $p < .001$; ** $p < .01$; * $p < .05$

$\chi^2 = 214.56$, $df = 20$, $CFI = .96$, $NFI = .95$, $GFI = .96$, $RMSEA = .09$

This model was tested based on the meta-analyzed correlation matrix among these variables (see Table 7). We excluded supervisory identification, supervisory commitment, and perceived supervisory support from this model because the inclusion of these three variables created empty cells in the correlation matrix.

OCBO = organizational citizenship behavior targeting the organization.

OCBS = organizational citizenship behavior targeting the supervisor.

Fig. 2. Target similarity model.

Table 8

Single indirect effects (with their bootstrapping bias-corrected confidence intervals) of organizationally-focused justice and supervisory-focused justice on OCBO and OCBS.

Variables	OCBO			OCBS		
	Indirect effect	95% CI	<i>p</i>	Indirect effect	95% CI	<i>p</i>
1. Organizationally-focused justice <i>via</i> perceived organizational support	.106***	(.079, .139)	.000	–		
2. Organizationally-focused justice <i>via</i> organizational identification	.028*	(.001, .055)	.043	–		
3. Organizationally-focused justice <i>via</i> affective organizational commitment	.031*	(.001, .060)	.045	–		
4. Organizationally-focused justice <i>via</i> organizational trust	.046**	(.016, .076)	.003	–		
5. Organizationally-focused justice <i>via</i> leader–member exchange	–			.051**	(.036, .070)	.001
6. Organizationally-focused justice <i>via</i> supervisory trust	–			.046**	(.031, .064)	.001
7. Supervisory-focused justice <i>via</i> leader–member exchange	–			.154**	(.117, .191)	.001
8. Supervisory-focused justice <i>via</i> supervisory trust	–			.120**	(.089, .154)	.001
9. Supervisory-focused justice <i>via</i> perceived organizational support	.065**	(.046, .088)	.001	–		
10. Supervisory-focused justice <i>via</i> organizational trust	.020**	(.007, .035)	.003	–		

Note. *** $p < .001$; ** $p < .01$; * $p < .05$.

the full test of the social exchange-mediated target similarity model), to an alternative model with justice types substituting for the source-based justice perception variables (i.e., no explicit matching of sources). This model showed significantly worse fit to data ($\chi^2 = 1825.39$, $df = 32$, $CFI = .71$, $NFI = .71$, $GFI = .82$, $RMSEA = .22$). Taken together, these combined results support Hypothesis 3.

In summary, our findings suggest that (a) multifoci justice represents a more powerful predictive framework as compared to justice type approaches that are less precise with regard to source in explaining variance in target-specific outcomes; (b) multifoci justice perceptions better predict target similar outcomes than target

dissimilar outcomes; (c) multifoci justice effects on target similar outcomes are mediated by source-specific social exchange.

Discussion

Summary

At its core, the field of behavioral ethics explores the normative ethicality of the actions taken by individuals, groups, and organizations (De Cremer, Mayer, & Schminke, 2010; Ferrell, Fraedrich, & Ferrell, 2008; Treviño et al., 2006). Whereas much of this research

takes the viewpoint of the actor (i.e., how and why individuals might engage in unethical behaviors; Aquino & Reed, 2002; Bandura, 1999; Blasi, 1984, 2005; Kohlberg, 1969; Rest, 1986), it has focused relatively less on phenomena surrounding employees' responses to the (un)ethical behavior of others (Greenbaum et al., 2011; Tenbrunsel & Smith-Crowe, 2008; Treviño et al., 2006). Given that this is a topic with which the organizational justice literature has been grappling for some time (and because reactions to injustice can often take the form of unethical behavior, such as CWB as explored herein), connecting the study of justice and ethics in organizations via a renewed focus on moral accountability seems apropos. As both of these literatures have focused *both* on accountability processes, *and* on compliance with normative rules (Folger & Cropanzano, 2001; Mikula et al., 1990; Treviño & Bies, 1997), and given the large number of researchers that conduct research spanning these two domains, we are surprised that multifoci approaches have not received more traction in the literature.

Thus, we set out to explore the extent to which past organizational justice research has been explicit about accountable parties. Further, by recoding the predictor–criterion relationships reported in past research according to the source that was referenced in the scales used, we sought to test whether improvement in prediction would be obtained by applying these two criteria (normative rules *and* an accountable party) to the prediction of source-specific outcomes. We sought to compare these target similarity effects to the effects of both traditional justice “types” (i.e., application of justice rules with multiple or unspecified sources), and to target dissimilar effects (i.e., multifoci justice perceptions predicting mismatched outcomes). Finally, we sought to test the extent to which target similarity effects are driven not only by the methodological alignment of the bandwidth and/or specificity of variables, but also by social exchange—the mechanism theorized to drive multifoci justice effects.

Our results indicated that, first, multifoci justice perceptions more strongly predicted outcomes directed at matched sources than did traditional type perceptions. We interpret this finding as being due to the fact that the multifoci justice variables harness *both* the power of normative justice rules *and* the explicit use of specific accountable parties in capturing employees' perceptions of justice. Second, we found that target similar effects were more robust than target dissimilar effects. This is likely due in part to precision gained methodologically by aligning the specificity or bandwidth of predictor and criterion variables. However, we also found target similar direct effects (e.g., the direct effect of organizationally-focused justice on OCBO) to be attenuated in the presence of target similar social exchange mediating variables. This speaks to the process by which source-specific justice engenders social exchange with that source, which further drives source-directed attitudes and behaviors.

Adding additional support to our hypotheses were findings that multifoci justice predicted variance in target similar outcomes that was over and above that accounted for by both the traditional justice types, and by target dissimilar effects. That being said, it is important to note that in 10 out of 18 cases, the most overall outcome variance was accounted for by virtue of including *both* target similar *and* target dissimilar justice into the regression model (i.e., including target dissimilar justice explained on average 0.8% additional variance in the outcomes over and above those explained by justice types and target similar justice). This was not unexpected, in that, as we mentioned at the start of the paper, target dissimilar effects have been both theorized and supported in past research (Lavelle et al., *in press*; Rupp & Cropanzano, 2002). The original conceptualization of target similarity is that cross-foci effects will exist, but be smaller in magnitude as compared to target similar effects. We found this to be the case in 16 out of 18 cases.

The two cases where target dissimilar effects were as strong as target similar effects were OCBO and CWBS. We do not have a

strong explanation as to why this may have been the case. One reason may have been that these two variables differed from the other variables in our meta-analytic dataset in that they represented clusters of related variables. As explained above, for parsimony, we categorized organization-directed civic virtue, sportsmanship, loyalty, extra-role behavior, and traditional OCBO (which often includes and collapses across these dimensions) as OCBO; and categorized supervisor-directed sabotage, retaliation, and aggression together as supervisor-directed CWB. It is possible that this was too broad of a brush stroke, and that differences between these related constructs confounded our results. Future research might consider exploring these issues in more depth.

Also, consistent with our expectations, we did detect target dissimilar effects to be more pronounced for supervisory-focused justice predicting organization-directed outcomes than for organizationally-focused justice predicting supervisor-directed outcomes. This further supports the particularly salient role played by the supervisor in influencing employees' attitudes and behaviors (Rupp & Cropanzano, 2002).

Comparison of current results with those of Colquitt et al. (2013)

As mentioned at the start of our paper, a secondary goal of our research was to sort through the issues that might explain why on one hand, the extant multifoci literature has consistently shown support for the multifoci/target similarity perspective, whereas on the other hand, Colquitt et al. (2013), on the basis of a recent meta-analytic investigation, concluded that there is little support for such an approach. Given that our meta-analysis, which was also based on several decades of empirical justice research, showed support for target similarity, we offer further comparison of these two meta-analytic studies in hopes of offering clarity to what may seem like completely divergent results.

Papers included in the meta-analysis

An initial point of comparison involves the sets of papers that served as input to these two meta-analytic investigations. Colquitt et al. (2013) reported identifying 413 papers (representing 493 independent samples), with the most recent published in 2010. In comparison, we identified 530 papers (representing 647 independent samples) for inclusion, with the most recent published in 2012. This might lead readers to conclude that our paper, with its larger *k* and *N*, and inclusion of papers published in 2011 and 2012, is more representative of the population and thus the inferences drawn from our findings are likely more accurate. While possible, we do not draw such a conclusion. Colquitt et al.'s meta-analytic dataset is large, covers a number of decades of research, and is for all intents and purposes, likely just as representative as ours. Consistent with arguments made in the methods literature, we doubt that this difference is the source of our divergent conclusions (see Aguinis, Dalton, Bosco, Pierce, & Dalton, 2011; Aguinis, Pierce, Bosco, Dalton, & Dalton, 2011).

Coding of papers according to source

Rather, where we think important differences between our study and Colquitt et al.'s lie is in the coding of the justice papers retrieved for analysis. As is described in our Methods section, we developed a detailed coding protocol with which to identify and code the source referenced in scales used to measure study variables. Often times the source was not clearly articulated in the paper itself, and thus our protocol dictated that we seek the original measures, and sometime contact original authors (of either the paper or scales), to be as sure as possible as to the source or sources referenced in justice and outcome items. This process revealed, as is illustrated in Fig. 1, that a large number of measures contained mixed or unspecified sources. We carefully excluded any

correlations stemming from such measures from our computation of the multifoci correlations, in that they would have contaminated our results and not provided a test of “pure” multifoci justice effects.

Colquitt et al. reported coding “the focus of the measure as either supervisor or organization when the items were clearly referenced to one of those sources,” and that “In some instances, the focus was unclear or varying, in which case the results were included in our overall summaries but not our focus breakdown” (p. 206). However, in reviewing their tables in light of their reported k and N , it appears that Colquitt et al. were much more liberal in their source coding than we were. That is, across the various justice and outcome relationships, Colquitt et al. were able to identify the justice source specified in prior studies, on average, about 90% of the time, and for many relationships (e.g., between PJ and OCB reported in their Table 2), 100% of the justice type variables were coded as referencing a source (either organization or supervisor). As is evident from our Fig. 1, this percentage is substantially higher than we were able to accomplish with our detailed coding protocol. In this case, we feel that our more conservative approach to coding the justice articles was essential in cleanly testing the robustness of the multifoci approach and the subsequent predictive power of the target similarity model. In contrast, the inclusion of any variables that were liberally coded to a source, where in reality the items suggest a mixed or ambiguous source, would contaminate the multifoci meta-analytic estimates, and reduce the likelihood of supporting the multifoci/target similarity model. We speculate that this difference in coding approaches is one of the major reasons that led Colquitt et al. (2013) to reach different conclusions from those reported here.

Creation of the multifoci justice variables

A second major difference between our paper and Colquitt et al.’s that should not be overlooked, involves the creation of the multifoci variables. This is a complex issue, in that the literature is yet to reach consensus on exactly how multifoci justice should be operationalized. As we explain in our paper and is articulated in more detail elsewhere (Lavelle et al., 2007, *in press*), the multifoci approach evolved from research by Malatesta and Byrne (1997) and Masterson et al. (2000), who made arguments (stemming from Bies & Moag, 1986) that employees attribute specific types of justice (e.g., interactional justice) to specific sources (e.g., the supervisor). Rupp and Cropanzano (2002) then extended this work, arguing and showing that employees are in fact able to differentiate multiple types of justice coming from multiple sources. This has been labeled a “fully-crossed” model, whereby all possible sources and types of justice are crossed.

However, more recent research has argued that employees can experience outcomes, procedures, and interactions from myriad sources, and over time, this event-level information is encoded and categorized into memory according to the sources held responsible for the treatment (regardless of whether the events involved distributive, procedural, or interactional elements) across events—leading to more global source-based justice perceptions that are aggregates of the justice types, all focused on a specific source (Guo, Rupp, Weiss, & Trougakos, 2011; Rupp, 2011; Rupp & Paddock, 2010).⁸ It is argued that the justice rule sets (distributive, procedural, and interactional) are used within and between events to hold various parties accountable for (un)just treatment.

⁸ In the spirit of full disclosure, it should be noted that our earlier work in this area (Liao & Rupp, 2005; Rupp & Cropanzano, 2002) originally set out to test for more global multifoci justice effects, where types were subsumed within source. It was only during the review process that we were pushed toward the fully-crossed model, due to the strong normative assumptions regarding justice types that were inherent to the field at that time.

This is how we conceptualized and operationalized multifoci justice in our paper, and consequently, how we created our multifoci justice variables for the meta-analyses. Thus, in our study, “organizationally-focused justice” contained all justice variables (regardless of type) where measures explicitly specified the organization as the source of justice, whereas “supervisory-focused justice” included all justice variables (regardless of type) where measures explicitly specified the supervisor as the source of justice. In this way, type is subsumed within source (and for the type variables, source is subsumed within type).

In contrast, Colquitt et al. (2013) employed a fully-crossed model, where each source of justice was considered meta-analytically for each type of justice. Thus, whereas we had two multifoci variables (organizationally-focused and supervisory-focused justice), Colquitt et al. had eight (i.e., two sources for four justice types). Fully crossing source and type has not only been criticized in the literature on the grounds of parsimony (Ambrose & Schminke, 2007; Rupp, Bashshur, & Liao, 2007), but also due to the reasons laid out in our paper—that the types of justice are not “types” at all, but rather sets of normative rules that individuals can use to evaluate the fairness of different types of events, but that ultimately require an accountable party (Colquitt et al., 2001).

Another way to think about these issues is to assess whether or not the source-based and type-based justice effects were compared in a balanced way (i.e., given equal footing in the comparison). For example, we compared procedural justice (PJ) effects to organizationally-focused justice (OJ) effects, where PJ effects included all reported PJ relationships (regardless of the source referenced in the PJ measures), and OJ effects included all organizationally-focused justice relationships (regardless of the type of justice referenced in the OJ measures). These are not the same comparisons made in the Colquitt et al. paper. In their case, continuing with the example above, Colquitt et al. (2013) compared the effects of PJ to the effects of organizationally-focused PJ and supervisory-focused PJ (or as another example, the effect of DJ to the effects of supervisory-focused DJ and organizationally-focused DJ). In this way, type and source are not given equal standing in the comparison, because one variable (e.g., organizationally-focused PJ) is actually just a sub-set of the variable to which it is being compared (e.g., PJ). These sorts of comparisons create a natural disadvantage in the estimation of source-based justice effects.

For these reasons, we feel that our method of creating the multifoci justice variables, and our subsequent results, are more relevant for understanding issues of multifoci justice and target similarity than those reported in Colquitt et al. (2013).

Creation of CWB variables

We should also note here that there were additional differences with regard to the creation of the CWB variables between our study and Colquitt et al.’s. Congruent with the literature (e.g., Griffin & Lopez, 2005), we classified active, destructive behaviors as CWB. This included variables such as retaliation, aggression, and deviant behaviors. In contrast, Colquitt et al. (2013) classified variables as CWB not only if they reflected “active deviance or retaliation” (p. 206), but also if they referenced “more passive withdrawal-style behaviors (e.g., absenteeism, tardiness)” (p. 206). Thus, there is a possibility that our differential findings with regard to CWB could be due to different sets of sub-constructs feeding this variable in the analyses.

Analytic strategies

A third important difference between our study and Colquitt et al.’s [as well as between Colquitt et al. (2001) and Colquitt et al. (2013)], which we mentioned at the start of the paper, involved the analytical approach taken. In Colquitt et al. (2013), the authors set up arguments and hypotheses to compare the

aggregated zero-order correlations between organizationally- vs. supervisory-focused DJ, PJ, and IJ and source-coded outcomes (their Hypotheses 3 and 4). Within each justice type, they calculated the average correlations between organizationally-focused justice and target-specific justice outcomes and compared them with the averaged correlations between supervisory-focused justice and the same outcomes. Thus, for each justice type separately, they considered whether the zero-order correlation between a justice facet and a target-similar outcome was larger than the zero-order correlation between the same facet with a target-dissimilar outcome (or stated differently, between an organizationally- vs. supervisory-focused facet and a target-specific outcome; e.g., the average correlation between OrgPJ and OCBO vs. the average correlation between SupPJ and OCBO). Based on the overlapping confidence intervals across the majority of these relationships, the authors concluded that matching the foci of justice with the target of the outcome did not yield stronger effect sizes (although in a few cases the results were consistent with what target similarity would suggest; i.e., matching the foci and target of justice and outcomes resulted in larger effect sizes). That said, their findings did not contradict target similarity by showing that target dissimilar effects were stronger than target similar effects.

In our meta-analysis, we attempted to compare the incremental predictive validity of multifoci justice perceptions (as source-based justice perceptions within which the types are combined), over type-based justice perceptions (within which sources are unspecified or combined), on target-specific justice outcomes. We set up hypotheses to explore if multifoci justice perceptions more strongly predicted target similar outcomes than do type-based justice perceptions. Based on the meta-analytic correlation matrices (each correlation coefficient in the matrix is the averaged relationship between variables of interest), we performed *multiple regression analyses* (modeled after the approach taken by Colquitt et al., 2001 for testing the system-agent model; see also Podsakoff, MacKenzie, & Bommer, 1996; Schmidt, Hunter, & Outerbridge, 1986) to examine if multifoci justice perceptions more strongly predicted target similar outcomes as compared to type-based justice perceptions. Doing so allowed us to isolate the predictive power of justice source (organizational and supervisory) in the presence of type-based justice perceptions. Our findings in general (with two exceptions) supported our hypotheses. Given our hypotheses also focused on the differential effects of justice constructs (sources and types), the use of a regression-based approach is preferable to comparing zero-order, bivariate relationships. Because multiple regression tells us about the relationship between each predictor and a criterion in the presence of other predictors, it is a useful tool for identifying the best combination of predictors in accounting for criterion variance (Mertler & Vannatta, 2005).

Overall purpose of the two studies

A final important difference between our paper and Colquitt et al.'s (2013), which can have a non-negligible influence on reported outcomes, is the overall purpose of the two respective studies. Our paper was specifically focused on theory testing, within a narrow and specific domain (i.e., multifoci justice), within which we have been working for over a decade. That is, we sought to explore the importance of specifying an accountable source in justice research, and to test the target similarity model in particular. Our approach to identifying and coding primary studies, as well as our analytic strategy, were informed very specifically by theory within this domain.

In contrast, the Colquitt et al. paper had a broader set of goals, focused on updating their prior (2001) meta-analysis according to a variety of theoretical developments that have gained strength in the literature (e.g., social exchange, the role of affect, entity vs. event distinctions, etc.). Colquitt et al. (2013) specifically defined

“justice” in their paper as “distributive, procedural, interpersonal, and informational justice” (see their Hypothesis 1, p. 201), and hence in no way took on the issue of “type” (and the best placement of justice rule violation within the structure of the justice construct) as we have in our paper. Colquitt et al. compared the effects of organizationally-focused justice vs. supervisory-focused justice (within each justice type) on target-specific outcomes (e.g., LMX, POS; see their Hypotheses 3 and 4). They did not directly compare the relative predictive validity of type-based justice perceptions vs. source-based justice perceptions on target-specific outcomes. As such, their study is less focused on exploring the optimal structure of the justice construct in accounting for variance in justice reactions, or on testing any specific theory in depth; and more focused on providing a higher-level summary of a broad range of inter-relationships (e.g., entity vs. event). This approach likely informed the authors’ decision to take a relatively more liberal approach to coding variables by source, to focus on zero-order, bivariate relationships as opposed to regression-based models, and to forgo creating multifoci composite variables, where types were subsumed within sources.

LeBreton, DeSimone, Scherer, Lee, and James (2013) speak in detail about how breadth vs. depth in focus, as well as substantive experience within a theoretical domain can have a marked effect on the judgment calls made in setting up and carrying out meta-analytic investigations. Further, scholars have repeatedly pointed out how differences in such judgment calls can lead to differential sets of findings and conclusions (e.g., Bobko & Roth, 2008; Bullock & Svyantek, 1985; Wanous, Sullivan, & Malinak, 1989). We see Colquitt et al.'s (2013) work as providing a broad summary that will no doubt inform future research in a number of ways. However, with regard to the explicit testing of the multifoci approach and target similarity model, we feel the results stemming from our specifically focused and theoretically informed investigation, where theory dictated the choice of and coding of variables, as well as the specific analytical approach taken, provide a more definitive test of theory compared to the more limited set of information provided by Colquitt et al. (2013) pertinent to this particular research question.

We hope that this comparison also highlights to readers that meta-analytic investigations simply cannot be viewed as definitive summaries and are in no way a substitute for deep theoretical thinking and continued (primary) empirical investigation (Bobko & Stone-Romero, 1998). Our hope is that this dialog will spawn more quality thinking and primary research within this domain.

Limitations and further research needs

Unsupported findings

The present study allowed for a large-scale investigation of our research questions via the incorporation of the empirical organizational justice literature, multiple coders with high inter-rater reliability, and the use of advanced meta-analysis, path analysis, and mediation techniques. Although the majority of our statistical tests produced effects that were consistent with our hypotheses, this was not the case 100% of the time. In particular, there were two unsupported aspects of our results that require further attention. The first involved predictions of OCBO (and to some degree, CWBS). For OCBO, we did not find organizationally-focused justice perceptions to out-predict either the justice types or supervisory-focused justice. For CWBS, we did not find supervisory-focused justice to predict as much incremental variance beyond type perceptions as did organizationally-focused justice perceptions. Although we cannot say with certainty why these particular results did not confirm our hypotheses, as we have explained above, these two variables differed from the other justice reactions in our dataset because they represented clusters of related variables. It is possible

that differences between these related constructs in these clusters of variables have confounded our results.

It is interesting to note that there have been some inconsistencies in the multifoci literature regarding target (dis)similar effects on OCBO (e.g., Rupp & Cropanzano, 2002 vs. Masterson et al., 2000). Lavelle et al. (2007) speculated that this may be due to the differences in measures used across studies. For example, Moorman and Blakely (1995) measure explicitly refers to behaviors directed toward the organization itself (e.g., “I defend the organization when other employees criticize it”), whereas Williams and Anderson’s (1991) items look more like general conscientiousness than organization-directed citizenship (e.g., attendance; giving advance notice when unable to come to work). Organ (1988) has argued that some of these behaviors, although theoretically organization-directed, may be reported on without an explicit beneficiary in mind. Future research might consider exploring these issues in more depth.

A second set of unsupported findings involved the incremental validity of source over type (vs. type over source) in the prediction of affective organizational commitment and supervisory trust. While we don’t have a clear explanation regarding why the results involving these specific variables did not support our hypotheses, some *post hoc* analyses did provide us with some preliminary ideas. Specifically, we found that among the 63 studies that provided correlations between PJ and affective organizational commitment, over 50% of them specified the organization as the source of procedural justice. Thus the averaged correlation between PJ and affective commitment ($\bar{r} = .40$, see Table 2) was calculated based on more target similar than target dissimilar effects in this case. A similar pattern was revealed for supervisory trust. Here, we found that among the 24 studies that provided correlations between IJ and supervisory trust, 19 of them specified the supervisor as the justice source. Thus, it is possible that the three justice types were able to explain as much or more incremental variance in these two particular outcomes over and above justice sources than vice versa due to some of the type correlations capitalizing on the target similar effects present in the data matrix. In summary, despite these two sets of unsupported findings, given that our predictions were supported across 16 of the 18 criterion variables, we believe that these exceptions did not invalidate our overall theoretical assertions.

Common method bias and heterogeneity among effect sizes

First, many of the outcome variables incorporated into our analyses were measured via self-report. Since justice is always a self-report variable (as it is perceptual in nature), our averaged correlations might be inflated due to common source bias. That being said, researchers have argued that despite common method bias in any single empirical study being a concern, the base rate of CMV is not very high (Doty & Glick, 1998; Spector, 2006). Further, many of the outcome variables involving performance and OCB were measured (in some primary studies) via supervisory-report.

Despite this, as an extra precaution, we performed additional analyses in order to examine if CMV may have inflated the averaged correlations between our variables of interest. Specifically, we compared the averaged correlation between justice perceptions and outcomes when the outcome was reported by the participants themselves vs. their supervisors or co-workers. For example, we compared the averaged correlation between interactional justice and OCBO when OCBO was reported by the participants themselves [$\bar{r} = .21, 95\%CI = (.13, .28), k = 13, N = 4174$] to the averaged correlation between interactional justice and OCBO when OCBO was reported by supervisors or co-workers [$\bar{r} = .20, 95\%CI = (.17, .23), k = 22, N = 5786$]. Comparing these averaged correlations did not reveal differences according to whether outcome variables were measured via self- vs. other-report, implying that CMV did not bias the averaged correlations

between variables. In short, we believe that the power gained by our meta-analytic technique (e.g., corrections for sampling error) and functional sample size address this issue to some extent and that CMV did not substantially bias our findings.

Second, we detected some degree of heterogeneity amongst our meta-analytic effect sizes. This is understandable given the large number of samples that were included in our meta-analysis (spanning nearly 30 years of research). However, this might also be reflective of moderators at work. Because our study was specifically focused on exploring relative criterion-related validities of type- vs. source-based justice perceptions, as well as the robustness of the target similarity model, we neither hypothesized nor tested for moderation effects. This may, however, be an item for consideration in future research.

Unspecified/mixed sources

It could also potentially be argued that the number of studies having mixed or unspecified sources may have to some extent impacted our results. That is, any correlation reported in a primary study using a justice measure with a mixed or unspecified source would not have appeared among our justice source correlations, whereas any measures referencing outcomes, procedures, or interactions, were included into the DJ, PJ, and/or IJ calculations regardless of whether the source was supervisor, organization, mixed, or unspecified. This means that a) the multifoci justice correlations were computed using a smaller number of studies (and thus a smaller number of participants) as compared to the justice type correlations, and b) the justice type correlations contain target similar, target dissimilar, and ambiguous correlations. For the justice measures that specified no source, study participants were free to conjure different sources as they responded to generalized statements about the adherence to (or violation of) justice rules. If different sources are conjured by different individuals, then the target similarity perspective would argue that correlations with outcomes would be attenuated in that predictors and outcomes would be aligned for some participants and not others. The same logic applies to measures referencing mixed sources, in that, unless the outcome variable was equally mixed (none were), all effects would be at least partially target dissimilar.

This is exactly what we set out to explore in this study, and indeed, our findings have to some extent supported this notion, by showing that multifoci, target similar justice effects were the most robust. Further, our results showed that although psychometrically distinguishable, the different sets of normative justice rules do not necessarily need to be distinguished from one another to provide optimal prediction of source-directed outcomes. Rather, predictive utility is gained by incorporating multiple sets of normative rules, specific to a source, into the prediction of target similar outcomes. To be clear, although our results show that justice types matter, as does justice from sources other than the one to which outcomes are referenced, our findings suggest that the prediction of employee attitudes and behaviors is best achieved by considering multiple sources of justice, and allowing the normative justice rules (i.e., justice types), to be freely subsumed within each source.

We might also argue that we have created a fairly conservative test of our model by comparing multifoci effects to the effects of type-based justice perceptions on target-specific outcomes, because the effects of the justice type estimates include both target similar and dissimilar correlations. Thus, the effect sizes for type-based justice perceptions (from multiple sources) on target-specific outcomes are in some ways capitalizing on the target similar correlations present in our database. However, our results revealed that, even under these conservative conditions (i.e., a smaller sample for the source-based estimates, and type-based estimates containing target similar correlations), a (pure) source-based approach was still found to be relatively superior for mapping the variance in

outcomes attributed to fairness perceptions. We should also note that even though the source-based estimates may have had relatively less power due to the relatively smaller set of studies (k , and thus less participants, N), the k and N for multifoci justice effect size estimates were still very large and most likely well representative of the current literature (see Fig. 1).

Other sources beyond the supervisor and organization

A final limitation lies in the fact that supervisors and organizations are certainly not the only sources referenced in justice measures to date, as recent research is beginning to study the effects of customer (Rupp & Spencer, 2006; Skarlicki et al., 2008) and co-worker injustice (Donovan et al., 1998; Li & Cropanzano, 2009). Unfortunately, there were not enough of these types of studies to be included in our meta-analyses. However, we hope this body of research will continue to amass studies such as these, so that these additional sources can be included in future meta-analytic investigations.

Theoretical contributions and future research directions

Our paper makes a number of important theoretical contributions. The impetus of our study was an observation that what have become generally accepted as the “types” of justice (i.e., DJ, PJ, and IJ), may represent an incomplete mapping of justice perceptions as a construct. This is because, first, the types refer to sets of normative rules that are theoretically used when forming a justice perception (and therefore antecedents to justice perceptions as opposed to the perceptions themselves; Ambrose & Arnaud, 2005). Secondly, these type-based perceptions, in absence of a specified source, lack the second agreed upon “ingredient” of a justice perception, namely, an identifiable party held accountable for the upholding or violating of the normative justice rules (Folger & Cropanzano, 2001). Finally, if these justice types are used to predict general or target dissimilar outcomes, this is again inconsistent with justice theory, which speaks very clearly about reactions to justice being targeted at the accountable party (Greenbaum et al., 2011). By showing empirically that, when we allow for all three of these ingredients to drive justice perceptions (i.e., normative rules, an accountable party, and a matched target), we predict more variance in outcomes than operationalizations containing only some of these perceptual ingredients (i.e., justice types, target dissimilar outcomes), we contribute both to the pushing forward of justice theory, and to solidifying the connections between justice and ethics research.

Considering the phenomenological experience of justice

To be clear, our goal is not to discredit the long and important history of distributive, procedural, and interactional justice in any way. Indeed, the research within these areas constitutes some of the most important contributions to the organizational behavior literature to date. In addition, the contributions of Colquitt's meta-analysis and measure (Colquitt, 2001; Colquitt et al., 2001) brought consensus and standardization to the field that was much needed at the time. Our only goal is to point out that, although we have been talking about outcomes, procedures, and interactions as the natural components of justice perceptions, (a) this was not how the constructs emerged theoretically, and (b) moral accountability has played a minor and secondary role in a large proportion of this research. Thus, in order to align how we conceptualize employee justice (employee experience with the compliance/violation of normative ethical rules in the process of holding parties accountable for their actions) with how we operationalize justice perceptions, our results suggest that future research should provide employees with perceptual targets with which to report on their experiences—parties to be potentially held accountable for (un)just behaviors.

That being said, if one's goal is simply the prediction of general attitudes and behaviors, there is plenty of research evidencing the utility of general (non-source-specific) type-based justice perceptions. In fact, following the bandwidth-fidelity arguments reviewed in this paper, global justice perceptions that are not even specific to the normative justice rules, but are rather predicted by them, would be an even more proximal route for predicting global attitudes and behaviors (Ambrose & Schminke, 2009). However, if one's interest is more person-centric in nature (Weiss & Rupp, 2011); that is, if one's focus is on understanding individuals' subjective experiences of justice (Gilliland, 2008; Guo et al., 2011; Rupp & Aquino, 2009); which would necessarily begin at the point when employees have encounters with others, are socialized into the culture and climate of the organization, and begin to form exchange relationships (for better or for worse) with networks of others over the course of their work experience; then considering issues related to norms, accountability, and targeted reactions seems quite relevant. Although the results presented herein cannot speak to the relative salience of these components as they naturally occur in the minds of employees over time, future research might certainly explore such issues.

For example, Rupp and Paddock (2010) have proposed a temporal model of information aggregation and judgment that argues for the incorporation of time into the study of justice phenomena, as well as the treatment of justice as an experiential phenomenon that involves the sequential encoding, categorizing, retrieval, and re-appraisal as employees experience successive events across the course of their working lives (see also Guo et al., 2011; Rupp, 2011). These authors posit that distributive, procedural, and interpersonal information comes into play at the level of the event. These events catalyze quick, automatic, and emotional (deontic) responses, which are then processed cognitively as individuals form initial fairness perceptions, which are only validated if a party can be identified as responsible for what has occurred (Folger & Cropanzano, 2001). Once this has occurred, the experience is tagged according to the source of (in)justice and encoded into memory using source as a categorization scheme. Over time, these memorized source-referenced justice events are aggregated to form individual-level perceptions of multifoci justice, which, with the further passing of time, are aggregated to form individual-level perceptions of overall justice (Ambrose & Schminke, 2009), and eventually shared perceptions of justice within working groups (i.e., justice climate, Rupp & Thornton, in press).

What this model implies is that how we choose to measure justice should largely depend on when we measure it and what stage in this dynamic process we are interested in exploring. In other words, measuring justice according to the meeting of normative criteria (Colquitt, 2001), according to an accountability framework (Rupp & Cropanzano, 2002), and as an overall, aggregate perceptual construct (Jones & Martens, 2009), might each be an appropriate approach, depending on when in the judgment/reaction process the researcher focuses his/her attention. This is an extremely ripe area for future research, and we hope the justice community can work together to further sort out these issues. We are encouraged by research that is beginning to incorporate time into hypotheses about justice perceptions (e.g., Barsky, Kaplan, & Beal, 2011; Holtz & Harold, 2009). We also think that simultaneously recognizing the complexity of the formation of justice perceptions over time and adding to the precision with which we measure such constructs will allow for research that further connects justice to the study of behavioral ethics (see Treviño et al., 2006 for a discussion of existing connections).

Social exchange theory

A second theoretical contribution is made by our modeling of social exchange as mediating the effects of multifoci justice on target similar outcomes – a finding that contributes to justice, eth-

ics, and social exchange theories. Indeed, the process by which justice engenders social exchange between employees and the parties with whom they work has been well articulated in the theoretical literature, and the relationships between justice and various social exchange variables (e.g., commitment, trust, support) have been offered in the empirical literature (see [Cropanzano & Mitchell, 2005](#) for a review). Our study serves as the first attempt at aggregating these findings meta-analytically in order to understand more fully the strength of the social exchange mechanism. As compared to the individual empirical studies, we modeled multiple social exchange variables simultaneously and aggregated numerous previously reported zero-order coefficients among variables to construct a large correlation matrix with which to examine the mediating role of social exchange. Specifically, our findings revealed that the total indirect effect of organizationally-focused justice perceptions on OCBO via all the social exchange variables (i.e., organizational trust, commitment, support, and identification) was .21, which is approaching a medium effect size based on [Cohen \(1988\)](#) criteria. Additionally, the total indirect effect of supervisory-focused justice perceptions on OCBS via all the social exchange variables (i.e., supervisory trust and LMX) was .28 (also medium in size based on Cohen's standards). Future research is needed that incorporates additional differentiated exchange relationships (e.g., with clients, customers, and co-workers), the interaction of these different exchange relationships on worker behavior and well-being, and the effects on employees over time, as relationships change, both due to episodic events, and due to individuals moving in and out of specified roles (e.g., a supervisor is promoted and is replaced by someone else).

Third, while not the focus of our present study, our findings did reveal that the single indirect effects of the two justice sources on OCBO/OCBS via different social exchange mediators varied in terms of their magnitude. This suggests that some social exchange variables (e.g., POS in the relationship between organizationally-focused justice and OCBO) might be more proximal mediating mechanisms as compared to other weaker (more distal) mediating mechanisms (e.g., organizational identification in the relationship between organizationally-focused justice and OCBO). It is possible that mediation may be occurring at multiple stages (see [Lavelle et al., 2007](#)), or that some mediators are more influential than other mediators. Future research might use these meta-analytic findings inductively to build more complex models of multifoci social exchange processes.

Also, our study revealed that the magnitude of ΔR^2 (i.e., the incremental variance explained above and beyond justice types) when supervisory justice was the focal source predicting outcomes targeting the supervisor was typically greater than the magnitude of ΔR^2 when organizationally-focused justice was the focal source predicting outcomes targeting the organization. Past research has argued that while employees are likely to hold supervisors and organizations accountable for (un)fair treatment received, employees may perceive supervisors as having more discretion in their decisions and behaviors as compared to organizations ([Folger & Cropanzano, 1998](#)). This would imply the engenderment of social exchange with different parties may differ by the source, and further validates the particularly important role of the supervisor in impacting employee attitudes and behaviors. Future research should explore how accountability processes may differ across various justice sources, due to differences in the employees' ability to attribute blame and assign responsibility ([Greenbaum et al., 2011](#)), and consequently, how this impacts subsequent exchange relationships with these various sources (e.g., co-workers, clients).

Target similarity

Moreover, our study is among the first to empirically synthesize prior justice research to examine the target similarity effects be-

tween workplace fairness and employee behaviors from two major justice sources simultaneously and in the presence of target dissimilar effects. Our findings support the target similarity model proposed by [Lavelle et al. \(2007\)](#) and offer new insights for future research. For example, as we stated above, more research on parties beyond the supervisor and organization are needed. In addition to this, future research might consider the moderators of target similarity effects. For example, power distance between employees and the foci of justice may influence the strength of justice reactions, which could be further moderated by the extent that power distance is valued among employees (e.g., [Lam, Schaubroeck, & Aryee, 2002](#); see [Triandis, Bontempo, Villareal, Asai, & Lucca, 1988](#) for ideas of how this might occur with regard to social exchange).

Also, recent research has speculated as to the possibility of joint (interactive) effects of multifoci justice variables on outcomes. For example, [Price, Lavelle, Henley, Cocchiara, and Buchanan \(2006\)](#) found that the voice provided by different parties at different stages of a decision process jointly influenced subsequent fairness perceptions, such that it only took one party acting unfairly at any stage in the process to influence reactions. [Lavelle, McMahan, and Harris \(2011\)](#), based on conservation of resources theory, predicted and found that injustice from another source weakened the effect of workgroup-focused justice perceptions and OCBs directed at the work group. Considering multifoci justice at the group level of analysis, [Rupp et al. \(2007\)](#) presented findings evidencing an interaction between supervisory- and organizationally-focused justice climate. They showed that the impact of inconsistent treatment from supervisors and the organization had a more detrimental effect on OCB than did consistently unfair treatment from both sources. Whereas the possibility of multifoci interactive effects is intriguing and could explain some of the cross-foci effects detected by our analyses, we were unfortunately unable to test such interactive effects within the context of our paper, in that such tests would require more information than what was reported in the primary studies feeding our meta-analytic database (see [Jiang, Lepak, Hu, & Baer, 2012](#); [Kelava et al., 2011](#)). As such, the literature would benefit from more primary research in this area.

Practical implications

Our research also has important implications for managing and growing both justice and ethics in the workplace. The most obvious implication involves training. Whereas ethics training has become increasingly common in organizations, what is much less common are training programs with the explicit goal of improving the level of fairness perceived by the workforce. What has been reported on in the literature is reflective of the history of justice research, in that it has been structured around the normative rules of distributive, procedural, and interactional justice.

For example, [Skarlicki and Latham \(1996, 1997\)](#) trained union leaders on the normative justice rules of procedural and interactional justice, and tracked improvement in union members' procedural and interactional justice perceptions. Similarly, [Rupp, Baldwin, and Bashshur \(2006\)](#) described a managerial developmental assessment center program that incorporated justice as a leadership competency. In this program, simulations, feedback, and follow-up 360 ratings were all modeled around the three types of justice (DJ, PJ, and IJ). All three of these studies showed that training managers on the justice rules can positively impact employees' perceptions of the extent to which managers uphold these criteria.

Our results suggest that the utility of training aimed at increasing justice in the workplace might be further maximized by taking an approach to training that focuses not only on the normative criteria for justice, but also accountability processes. That is, justice

training might take a multi-stakeholder approach, aimed at identifying the various exchange relationships with which employees may be engaged, in order to design practices that promote fairness in the workplace through these complex social networks (Shapiro, Brass, & Labianca, 2008).

Such an approach might be integrated into existing ethics training programs, which typically focus on key company values, ethical issues specific to the trainees' work context, the resolving of ethical dilemmas via ethical principles, managing competing pressures that create ethical challenges, legal issues, and whistleblowing policies (Valentine & Fleischman, 2004). Given that the goals of ethics and justice training programs would both involve multiple stakeholders, the management of multiple (sometime conflicting) exchange relationships, and the evaluation of behaviors through a lens of moral norms and accountability, this seems like a natural marriage. Furthermore, the ethics training literature offers many ways in which justice interventions might be made more effective. This includes involving top leadership, including employees at all levels, and incorporating justice into performance appraisal practices (White & Lam, 2000). Whereas moral accountability is an issue that bridges issues of justice and ethics, this is another way these fields could be more thoroughly integrated, through practice (see also Greenberg, 2009; Rupp & Aquino, 2009).

In addition, because social capital is built up overtime (Nahapiet & Ghoshal, 1998), and because of the long-term time horizon of social exchange relationships (Blau, 1964; Konovsky & Pugh, 1994), it is important for organizational leadership to recognize the importance of ongoing assessment and maintenance of employees' myriad social ties. This speaks to the importance of employee surveys in organizations, and the potential for including fairness assessments into 360-degree/multirater feedback programs. Further, although we did find strong evidence for target similarity effects, we did find some cross-foci effects, where fairness perceptions about one source predicted outcome directed at another source. This again speaks to the importance of strategic interventions focused on ensuring that employees are treated consistently fairly, by as many sources as possible (see Rupp et al., 2007). Finally, as discussed above, our findings highlighted the particularly important role of supervisor fairness and social exchange in influencing myriad outcomes. This points to the importance of supportive supervisor–employee relations, and suggests that supervisors should likely play a central role in the promotion of justice in the workplace (Folger & Cropanzano, 1998).

Conclusion

Although our traditional sets of normative justice rules are indeed relevant for understanding employees' attitudes and behaviors, integrating these criteria with an explicit, morally accountable source increases our ability to more fully map the perceptual domain of employee justice. As argued by Trevino et al. (2006), behavioral ethics research has not paid enough attention to the consequences of ethical and/or unethical behaviors. Multifoci justice provides insights as to how recipients react toward others' (un)ethical behavior, how such treatment impacts recipients' exchange relationships with these actors, and how recipients' subsequent attitudes and behaviors are shaped by these source-specific exchange relationships. Our paper presents an up-to-date systematic and comprehensive synthesis of the empirical justice literature through the lens of multifoci justice. Guided by the prevailing theory in this area, and using meta-analysis in combination with regression/path-analysis, our research provides empirical support for this perspective. We hope that our research catalyzes future behavioral ethics research that considers the dynamic and complex networks within which employees must work, which breed many different forms of justice-related encounters.

Acknowledgments

The authors would like to thank Maureen Ambrose, James Lav-elle, Russell Cropanzano, Marcus Crede, and two anonymous reviewers for guidance and feedback on our work. We also thank Long He, Wonjoon Chung, Yan Zhou, J. Oliver Siy, Jooyeon Seok, Alek Reed, John Cukierski, Michelle Chapekis, and Haley Park for their research assistance. Finally, we acknowledge the financial support of the University of Illinois at Urbana-Champaign, City University of Hong Kong, and Purdue University, without which this project would not have been possible. The first two authors contributed equally to this research.

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